

Multiple Regression Analysis: OLS Asymptotics

In Chapters 3 and 4, we covered what are called *finite sample*, *small sample*, or *exact* properties of the OLS estimators in the population model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u. \quad [5.1]$$

For example, the unbiasedness of OLS (derived in Chapter 3) under the first four Gauss-Markov assumptions is a finite sample property because it holds for *any* sample size n (subject to the mild restriction that n must be at least as large as the total number of parameters in the regression model, $k + 1$). Similarly, the fact that OLS is the best linear unbiased estimator under the full set of Gauss-Markov assumptions (MLR.1 through MLR.5) is a finite sample property.

In Chapter 4, we added the classical linear model Assumption MLR.6, which states that the error term u is normally distributed and independent of the explanatory variables. This allowed us to derive the *exact* sampling distributions of the OLS estimators (conditional on the explanatory variables in the sample). In particular, Theorem 4.1 showed that the OLS estimators have normal sampling distributions, which led directly to the t and F distributions for t and F statistics. If the error is not normally distributed, the distribution of a t statistic is not exactly t , and an F statistic does not have an exact F distribution for any sample size.

In addition to finite sample properties, it is important to know the **asymptotic properties** or **large sample properties** of estimators and test statistics. These properties are not defined for a particular sample size; rather, they are defined as the sample size grows without bound. Fortunately, under the assumptions we have made, OLS has satisfactory large sample properties. One practically important finding is that even without the normality assumption (Assumption MLR.6), t and F statistics have *approximately* t and F distributions, at least in large sample sizes. We discuss this in more detail in Section 5.2, after we cover the consistency of OLS in Section 5.1.

Because the material in this chapter is more difficult to understand, and because one can conduct empirical work without a deep understanding of its contents, this chapter may be skipped. However, we will necessarily refer to large sample properties of OLS when we relax the homoskedasticity assumption in Chapter 8 and when we delve into estimation using time series data in Part 2. Furthermore, virtually all advanced econometric methods derive their justification using large-sample analysis, so readers who will continue into Part 3 should be familiar with the contents of this chapter.

5.1 Consistency

Unbiasedness of estimators, although important, cannot always be achieved. For example, as we discussed in Chapter 3, the standard error of the regression, $\hat{\sigma}$, is not an unbiased estimator for σ , the standard deviation of the error u in a multiple regression model. Although the OLS estimators are unbiased under MLR.1 through MLR.4, in Chapter 11 we will find that there are time series regressions where the OLS estimators are not unbiased. Further, in Part 3 of the text, we encounter several other estimators that are biased yet useful.

Although not all useful estimators are unbiased, virtually all economists agree that **consistency** is a minimal requirement for an estimator. The Nobel Prize-winning econometrician Clive W. J. Granger once remarked, “If you can’t get it right as n goes to infinity, you shouldn’t be in this business.” The implication is that, if your estimator of a particular population parameter is not consistent, then you are wasting your time.

There are a few different ways to describe consistency. Formal definitions and results are given in Appendix C; here, we focus on an intuitive understanding. For concreteness, let $\hat{\beta}_j$ be the OLS estimator of β_j for some j . For each n , $\hat{\beta}_j$ has a probability distribution (representing its possible values in different random samples of size n). Because $\hat{\beta}_j$ is unbiased under Assumptions MLR.1 through MLR.4, this distribution has mean value β_j . If this estimator is consistent, then the distribution of $\hat{\beta}_j$ becomes more and more tightly distributed around β_j as the sample size grows. As n tends to infinity, the distribution of $\hat{\beta}_j$ collapses to the single point β_j . In effect, this means that we can make our estimator arbitrarily close to β_j if we can collect as much data as we want. This convergence is illustrated in Figure 5.1.

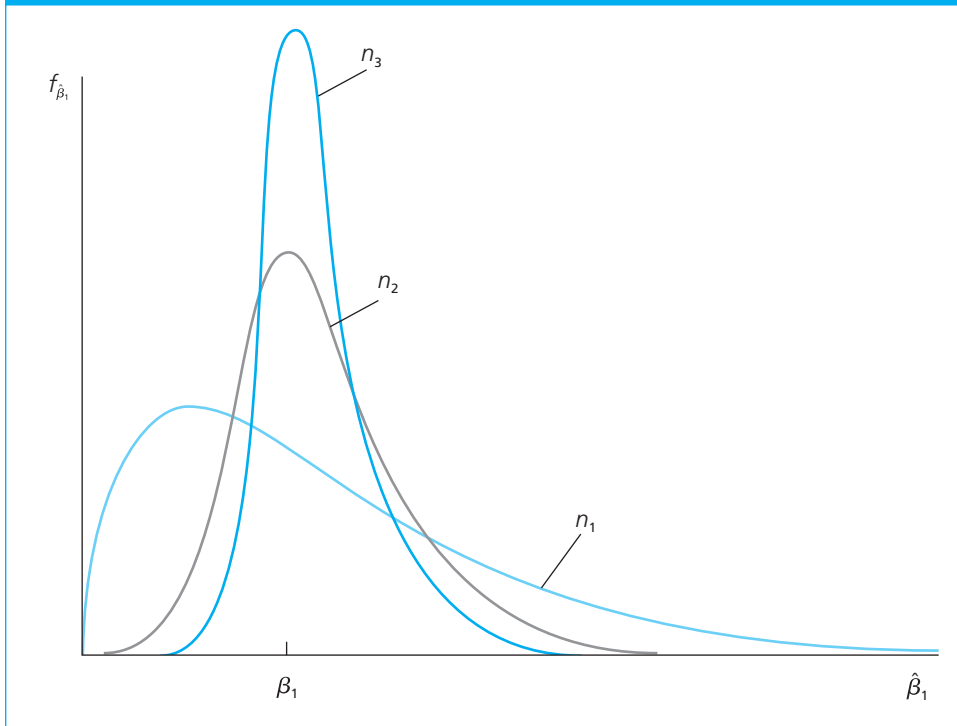
Naturally, for any application, we have a fixed sample size, which is a major reason an asymptotic property such as consistency can be difficult to grasp. Consistency involves a thought experiment about what would happen as the sample size gets large (while, at the same time, we obtain numerous random samples for each sample size). If obtaining more and more data does not generally get us closer to the parameter value of interest, then we are using a poor estimation procedure.

Conveniently, the same set of assumptions implies both unbiasedness and consistency of OLS. We summarize with a theorem.

THEOREM 5.1

CONSISTENCY OF OLS

Under Assumptions MLR.1 through MLR.4, the OLS estimator $\hat{\beta}_j$ is consistent for β_j , for all $j = 0, 1, \dots, k$.

FIGURE 5.1 Sampling distributions of $\hat{\beta}_1$ for sample sizes $n_1 < n_2 < n_3$.

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A general proof of this result is most easily developed using the matrix algebra methods described in Appendices D and E. But we can prove Theorem 5.1 without difficulty in the case of the simple regression model. We focus on the slope estimator, $\hat{\beta}_1$.

The proof starts out the same as the proof of unbiasedness: we write down the formula for $\hat{\beta}_1$, and then plug in $y_i = \beta_0 + \beta_1 x_{i1} + u_i$:

$$\begin{aligned}\hat{\beta}_1 &= \left(\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i \right) / \left(\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2 \right) \\ &= \beta_1 + \left(n^{-1} \sum_{i=1}^n (x_{i1} - \bar{x}_1) u_i \right) / \left(n^{-1} \sum_{i=1}^n (x_{i1} - \bar{x}_1)^2 \right),\end{aligned}\tag{5.2}$$

where dividing both the numerator and denominator by n does not change the expression but allows us to directly apply the law of large numbers. When we apply the law of large numbers to the averages in the second part of equation (5.2), we conclude that the numerator and denominator converge in probability to the population quantities, $\text{Cov}(x_1, u)$ and $\text{Var}(x_1)$, respectively. Provided that $\text{Var}(x_1) \neq 0$ —which is assumed in MLR.3—we can use the properties of *probability limits* (see Appendix C) to get

$$\begin{aligned}\text{plim } \hat{\beta}_1 &= \beta_1 + \text{Cov}(x_1, u) / \text{Var}(x_1) \\ &= \beta_1 \text{ because } \text{Cov}(x_1, u) = 0.\end{aligned}\tag{5.3}$$

We have used the fact, discussed in Chapters 2 and 3, that $E(u|x_1) = 0$ (Assumption MLR.4) implies that x_1 and u are uncorrelated (have zero covariance).

As a technical matter, to ensure that the probability limits exist, we should assume that $\text{Var}(x_1) < \infty$ and $\text{Var}(u) < \infty$ (which means that their probability distributions are not too spread out), but we will not worry about cases where these assumptions might fail.

The previous arguments, and equation (5.3) in particular, show that OLS is consistent in the simple regression case if we assume only zero correlation. This is also true in the general case. We now state this as an assumption.

Assumption MLR.4'

Zero Mean and Zero Correlation

$$E(u) = 0 \text{ and } \text{Cov}(x_j, u) = 0, \text{ for } j = 1, 2, \dots, k.$$

Assumption MLR.4' is weaker than Assumption MLR.4 in the sense that the latter implies the former. One way to characterize the zero conditional mean assumption, $E(u|x_1, \dots, x_k) = 0$, is that *any* function of the explanatory variables is uncorrelated with u . Assumption MLR.4' requires only that each x_j is uncorrelated with u (and that u has a zero mean in the population). In Chapter 2 we actually motivated the OLS estimator for simple regression using Assumption MLR.4', and the first order conditions for OLS in the multiple regression case, given in equations (3.13), are simply the sample analogs of the population zero correlation assumptions (and zero mean assumption). Therefore, in some ways, Assumption MLR.4' is more natural an assumption because it leads directly to the OLS estimates. Further, when we think about violations of Assumption MLR.4, we usually think in terms of $\text{Cov}(x_j, u) \neq 0$ for some j . So how come we have used Assumption MLR.4 until now? There are two reasons, both of which we have touched on earlier. First, OLS turns out to be biased (but consistent) under Assumption MLR.4' if $E(u|x_1, \dots, x_k)$ depends on any of the x_j . Because we have previously focused on finite sample, or exact, sampling properties of the OLS estimators, we have needed the stronger zero conditional mean assumption.

Second, and probably more important, is that the zero conditional mean assumption means that we have properly modeled the population regression function (PRF). That is, under Assumption MLR.4 we can write

$$E(y|x_1, \dots, x_k) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k,$$

and so we can obtain partial effects of the explanatory variables on the average or expected value of y . If we instead only assume Assumption MLR.4', $\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$ need not represent the population regression function, and we face the possibility that some nonlinear functions of the x_j , such as x_j^2 , could be correlated with the error u . A situation like this means that we have neglected nonlinearities in the model that could help us better explain y ; if we knew that, we would usually include such nonlinear functions. In other words, most of the time we hope to get a good estimate of the PRF, and so the zero conditional mean assumption is natural. Nevertheless, the weaker zero correlation assumption turns out to be useful in interpreting OLS estimation of a linear model as providing the best linear approximation to the PRF. It is also used in more advanced settings, such as in Chapter 15, where we have no interest in modeling a PRF. For further discussion of this somewhat subtle point, see Wooldridge (2010, Chapter 4).

Deriving the Inconsistency in OLS

Just as failure of $E(u|x_1, \dots, x_k) = 0$ causes bias in the OLS estimators, correlation between u and *any* of $x_1, x_2, \dots, x_k$ generally causes *all* of the OLS estimators to be inconsistent. This simple but important observation is often summarized as: *if the error is correlated with any of the independent variables, then OLS is biased and inconsistent*. This is very unfortunate because it means that any bias persists as the sample size grows.

In the simple regression case, we can obtain the inconsistency from the first part of equation (5.3), which holds whether or not u and x_1 are uncorrelated. The **inconsistency** in $\hat{\beta}_1$ (sometimes loosely called the **asymptotic bias**) is

$$\text{plim } \hat{\beta}_1 - \beta_1 = \text{Cov}(x_1, u) / \text{Var}(x_1). \quad [5.4]$$

Because $\text{Var}(x_1) > 0$, the inconsistency in $\hat{\beta}_1$ is positive if x_1 and u are positively correlated, and the inconsistency is negative if x_1 and u are negatively correlated. If the covariance between x_1 and u is small relative to the variance in x_1 , the inconsistency can be negligible; unfortunately, we cannot even estimate how big the covariance is because u is unobserved.

We can use (5.4) to derive the asymptotic analog of the omitted variable bias (see Table 3.2 in Chapter 3). Suppose the true model,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + v,$$

satisfies the first four Gauss-Markov assumptions. Then v has a zero mean and is uncorrelated with x_1 and x_2 . If $\hat{\beta}_0, \hat{\beta}_1$, and $\hat{\beta}_2$ denote the OLS estimators from the regression of y on x_1 and x_2 , then Theorem 5.1 implies that these estimators are consistent. If we omit x_2 from the regression and do the simple regression of y on x_1 , then $u = \beta_2 x_2 + v$. Let $\tilde{\beta}_1$ denote the simple regression slope estimator. Then

$$\text{plim } \tilde{\beta}_1 = \beta_1 + \beta_2 \delta_1, \quad [5.5]$$

where

$$\delta_1 = \text{Cov}(x_1, x_2) / \text{Var}(x_1). \quad [5.6]$$

Thus, for practical purposes, we can view the inconsistency as being the same as the bias. The difference is that the inconsistency is expressed in terms of the population variance of x_1 and the population covariance between x_1 and x_2 , while the bias is based on their sample counterparts (because we condition on the values of x_1 and x_2 in the sample).

If x_1 and x_2 are uncorrelated (in the population), then $\delta_1 = 0$, and $\tilde{\beta}_1$ is a consistent estimator of β_1 (although not necessarily unbiased). If x_2 has a positive partial effect on y , so that $\beta_2 > 0$, and x_1 and x_2 are positively correlated, so that $\delta_1 > 0$, then the inconsistency in $\tilde{\beta}_1$ is positive. And so on. We can obtain the direction of the inconsistency or asymptotic bias from Table 3.2. If the covariance between x_1 and x_2 is small relative to the variance of x_1 , the inconsistency can be small.

EXAMPLE 5.1

HOUSING PRICES AND DISTANCE FROM AN INCINERATOR

Let y denote the price of a house (*price*), let x_1 denote the distance from the house to a new trash incinerator (*distance*), and let x_2 denote the “quality” of the house (*quality*). The variable *quality* is left vague so that it can include things like size of the house and lot, number of bedrooms and bathrooms, and intangibles such as attractiveness of the neighborhood. If the incinerator depresses house prices, then β_1 should be positive: everything else being equal, a house that is farther away from the incinerator is worth more. By definition, β_2 is positive since higher quality houses sell for more, other factors being equal. If the incinerator was built farther away, on average, from better homes, then *distance* and *quality* are positively correlated, and so $\delta_1 > 0$. A simple regression of *price* on *distance* [or $\log(\text{price})$ on $\log(\text{distance})$] will tend to overestimate the effect of the incinerator: $\beta_1 + \beta_2\delta_1 > \beta_1$.

EXPLORING FURTHER 5.1

Suppose that the model

$$\text{score} = \beta_0 + \beta_1 \text{skipped} + \beta_2 \text{priGPA} + u$$

satisfies the first four Gauss-Markov assumptions, where *score* is score on a final exam, *skipped* is number of classes skipped, and *priGPA* is GPA prior to the current semester. If $\hat{\beta}_1$ is from the simple regression of *score* on *skipped*, what is the direction of the asymptotic bias in $\hat{\beta}_1$?

An important point about inconsistency in OLS estimators is that, by definition, the problem does not go away by adding more observations to the sample. If anything, the problem gets worse with more data: the OLS estimator gets closer and closer to $\beta_1 + \beta_2\delta_1$ as the sample size grows.

Deriving the sign and magnitude of the inconsistency in the general k regressor case is harder, just as deriving the bias is more difficult. We need to remember that if we have the model in equation (5.1)

where, say, x_1 is correlated with u but the other independent variables are uncorrelated with u , all of the OLS estimators are generally inconsistent. For example, in the $k = 2$ case,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

suppose that x_2 and u are uncorrelated but x_1 and u are correlated. Then the OLS estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ will generally both be inconsistent. (The intercept will also be inconsistent.) The inconsistency in $\hat{\beta}_2$ arises when x_1 and x_2 are correlated, as is usually the case. If x_1 and x_2 are uncorrelated, then any correlation between x_1 and u does *not* result in the inconsistency of $\hat{\beta}_2$: $\text{plim } \hat{\beta}_2 = \beta_2$. Further, the inconsistency in $\hat{\beta}_1$ is the same as in (5.4). The same statement holds in the general case: if x_1 is correlated with u , but x_1 and u are uncorrelated with the other independent variables, then only $\hat{\beta}_1$ is inconsistent, and the inconsistency is given by (5.4). The general case is very similar to the omitted variable case in Section 3A.4 of Appendix 3A.

5.2 Asymptotic Normality and Large Sample Inference

Consistency of an estimator is an important property, but it alone does not allow us to perform statistical inference. Simply knowing that the estimator is getting closer to the population value as the sample size grows does not allow us to test hypotheses about the parameters. For testing, we need the sampling distribution of the OLS estimators. Under the classical linear model assumptions MLR.1 through MLR.6, Theorem 4.1 shows that the sampling distributions are normal. This result is the basis for deriving the t and F distributions that we use so often in applied econometrics.

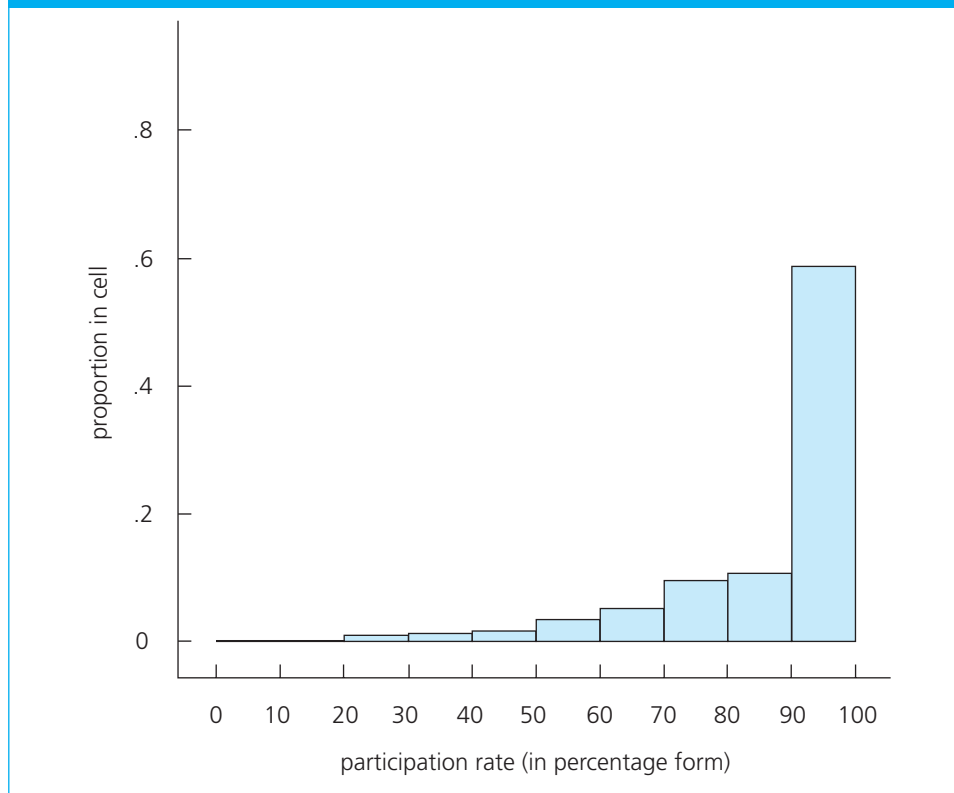
The exact normality of the OLS estimators hinges crucially on the normality of the distribution of the error, u , in the population. If the errors $u_1, u_2, \dots, u_n$ are random draws from some distribution other than the normal, the $\hat{\beta}_j$ will not be normally distributed, which means that the t statistics will not have t distributions and the F statistics will not have F distributions. This is a potentially serious problem because our inference hinges on being able to obtain critical values or p -values from the t or F distributions.

Recall that Assumption MLR.6 is equivalent to saying that the distribution of y given $x_1, x_2, \dots, x_k$ is normal. Because y is observed and u is not, in a particular application, it is much easier to think about whether the distribution of y is likely to be normal. In fact, we have already seen a few examples where y definitely cannot have a conditional normal distribution. A normally distributed random variable is symmetrically distributed about its mean, it can take on any positive or negative value (but with zero probability), and more than 95% of the area under the distribution is within two standard deviations.

In Example 3.5, we estimated a model explaining the number of arrests of young men during a particular year (*narr86*). In the population, most men are not arrested during the year, and the vast majority are arrested one time at the most. (In the sample of 2,725 men in the data set CRIME1.RAW, fewer than 8% were arrested more than once during 1986.) Because *narr86* takes on only two values for 92% of the sample, it cannot be close to being normally distributed in the population.

In Example 4.6, we estimated a model explaining participation percentages (*prate*) in 401(k) pension plans. The frequency distribution (also called a *histogram*) in Figure 5.2 shows

FIGURE 5.2 Histogram of *prate* using the data in 401K.RAW.



that the distribution of *prate* is heavily skewed to the right, rather than being normally distributed. In fact, over 40% of the observations on *prate* are at the value 100, indicating 100% participation. This violates the normality assumption even conditional on the explanatory variables.

We know that normality plays no role in the unbiasedness of OLS, nor does it affect the conclusion that OLS is the best linear unbiased estimator under the Gauss-Markov assumptions. But exact inference based on *t* and *F* statistics requires MLR.6. Does this mean that, in our analysis of *prate* in Example 4.6, we must abandon the *t* statistics for determining which variables are statistically significant? Fortunately, the answer to this question is *no*. Even though the y_i are not from a normal distribution, we can use the central limit theorem from Appendix C to conclude that the OLS estimators satisfy **asymptotic normality**, which means they are approximately normally distributed in large enough sample sizes.

THEOREM 5.2

ASYMPTOTIC NORMALITY OF OLS

Under the Gauss-Markov Assumptions MLR.1 through MLR.5,

- (i) $\sqrt{n}(\hat{\beta}_j - \beta_j) \overset{d}{\rightarrow} \text{Normal}(0, \sigma^2/a_j^2)$, where $\sigma^2/a_j^2 > 0$ is the **asymptotic variance** of $\sqrt{n}(\hat{\beta}_j - \beta_j)$; for the slope coefficients, $a_j^2 = \text{plim} \left(n^{-1} \sum_{i=1}^n \hat{r}_{ij}^2 \right)$, where the $\hat{r}_{ij}$ are the residuals from regressing x_j on the other independent variables. We say that $\hat{\beta}_j$ is *asymptotically normally distributed* (see Appendix C);
- (ii) $\hat{\sigma}^2$ is a consistent estimator of $\sigma^2 = \text{Var}(u)$;
- (iii) For each j ,

$$(\hat{\beta}_j - \beta_j)/\text{sd}(\hat{\beta}_j) \overset{d}{\rightarrow} \text{Normal}(0, 1)$$

and

$$(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j) \overset{d}{\rightarrow} \text{Normal}(0, 1), \quad [5.7]$$

where $\text{se}(\hat{\beta}_j)$ is the usual OLS standard error.

The proof of asymptotic normality is somewhat complicated and is sketched in the appendix for the simple regression case. Part (ii) follows from the law of large numbers, and part (iii) follows from parts (i) and (ii) and the asymptotic properties discussed in Appendix C.

Theorem 5.2 is useful because the normality assumption MLR.6 has been dropped; the only restriction on the distribution of the error is that it has finite variance, something we will always assume. We have also assumed zero conditional mean (MLR.4) and homoskedasticity of u (MLR.5).

In trying to understand the meaning of Theorem 5.2, it is important to keep separate the notions of the population distribution of the error term, u , and the sampling distributions of the $\hat{\beta}_j$ as the sample size grows. A common mistake is to think that something is happening to the distribution of u —namely, that it is getting “closer” to normal—as the sample size grows. But remember that the population distribution is immutable and has nothing to do with the sample size. For example, we previously discussed *narr86*, the number of

times a young man is arrested during the year 1986. The nature of this variable—it takes on small, nonnegative integer values—is fixed in the population. Whether we sample 10 men or 1,000 men from this population obviously has no effect on the population distribution.

What Theorem 5.2 says is that, regardless of the population distribution of u , the OLS estimators, when properly standardized, have approximate standard normal distributions. This approximation comes about by the central limit theorem because the OLS estimators involve—in a complicated way—the use of sample averages. Effectively, the sequence of distributions of averages of the underlying errors is approaching normality for virtually any population distribution.

Notice how the standardized $\hat{\beta}_j$ has an asymptotic standard normal distribution whether we divide the difference $\hat{\beta}_j - \beta_j$ by $\text{sd}(\hat{\beta}_j)$ (which we do not observe because it depends on σ) or by $\text{se}(\hat{\beta}_j)$ (which we can compute from our data because it depends on $\hat{\sigma}$). In other words, from an asymptotic point of view it does not matter that we have to replace σ with $\hat{\sigma}$. Of course, replacing σ with $\hat{\sigma}$ affects the exact distribution of the standardized $\hat{\beta}_j$. We just saw in Chapter 4 that under the classical linear model assumptions, $(\hat{\beta}_j - \beta_j)/\text{sd}(\hat{\beta}_j)$ has an exact $\text{Normal}(0,1)$ distribution and $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ has an exact t_{n-k-1} distribution.

How should we use the result in equation (5.7)? It may seem one consequence is that, if we are going to appeal to large-sample analysis, we should now use the standard normal distribution for inference rather than the t distribution. But from a practical perspective it is just as legitimate to write

$$(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j) \stackrel{\Delta}{=} t_{n-k-1} = t_{df}, \quad [5.8]$$

because t_{df} approaches the $\text{Normal}(0,1)$ distribution as df gets large. Because we know under the CLM the t_{n-k-1} holds exactly, it makes sense to treat $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ as a t_{n-k-1} random variable generally, even when MLR.6 does not hold.

Equation (5.8) tells us that t testing and the construction of confidence intervals are carried out *exactly* as under the classical linear model assumptions. This means that our analysis of dependent variables like *prate* and *narr86* does not have to change at all if the Gauss-Markov assumptions hold: in both cases, we have at least 1,500 observations, which is certainly enough to justify the approximation of the central limit theorem.

If the sample size is not very large, then the t distribution can be a poor approximation to the distribution of the t statistics when u is not normally distributed. Unfortunately, there are no general prescriptions on how big the sample size must be before the approximation is good enough. Some econometricians think that $n = 30$ is satisfactory, but this cannot be sufficient for all possible distributions of u . Depending on the distribution of u , more observations may be necessary before the central limit theorem delivers a useful approximation. Further, the quality of the approximation depends not just on n , but on the df , $n - k - 1$: With more independent variables in the model, a larger sample size is usually needed to use the t approximation. Methods for inference with small degrees of freedom and nonnormal errors are outside the scope of this text. We will simply use the t statistics as we always have without worrying about the normality assumption.

It is very important to see that Theorem 5.2 *does* require the homoskedasticity assumption (along with the zero conditional mean assumption). If $\text{Var}(y|\mathbf{x})$ is not constant, the usual t statistics and confidence intervals are invalid no matter how large the sample size is; the central limit theorem does not bail us out when it comes to heteroskedasticity. For this reason, we devote all of Chapter 8 to discussing what can be done in the presence of heteroskedasticity.

One conclusion of Theorem 5.2 is that $\hat{\sigma}^2$ is a consistent estimator of σ^2 ; we already know from Theorem 3.3 that $\hat{\sigma}^2$ is unbiased for σ^2 under the Gauss-Markov assumptions. The consistency implies that $\hat{\sigma}$ is a consistent estimator of σ , which is important in establishing the asymptotic normality result in equation (5.7).

Remember that $\hat{\sigma}$ appears in the standard error for each $\hat{\beta}_j$. In fact, the estimated variance of $\hat{\beta}_j$ is

$$\widehat{\text{Var}}(\hat{\beta}_j) = \frac{\hat{\sigma}^2}{\text{SST}_j(1 - R_j^2)}, \quad [5.9]$$

EXPLORING FURTHER 5.2

In a regression model with a large sample size, what is an approximate 95% confidence interval for $\hat{\beta}_j$ under MLR.1 through MLR.5? We call this an **asymptotic confidence interval**.

where SST_j is the total sum of squares of x_j in the sample, and R_j^2 is the R -squared from regressing x_j on all of the other independent variables. In Section 3.4, we studied each component of (5.9), which we will now expound on in the context of asymptotic analysis. As the sample size grows, $\hat{\sigma}^2$ converges in probability to the

constant σ^2 . Further, R_j^2 approaches a number strictly between zero and unity (so that $1 - R_j^2$ converges to some number between zero and one). The sample variance of x_j is SST_j/n , and so SST_j/n converges to $\text{Var}(x_j)$ as the sample size grows. This means that SST_j grows at approximately the same rate as the sample size: $\text{SST}_j \approx n\sigma_j^2$, where σ_j^2 is the population variance of x_j . When we combine these facts, we find that $\widehat{\text{Var}}(\hat{\beta}_j)$ shrinks to zero at the rate of $1/n$; this is why larger sample sizes are better.

When u is not normally distributed, the square root of (5.9) is sometimes called the **asymptotic standard error**, and t statistics are called **asymptotic t statistics**. Because these are the same quantities we dealt with in Chapter 4, we will just call them standard errors and t statistics, with the understanding that sometimes they have only large-sample justification. A similar comment holds for an **asymptotic confidence interval** constructed from the asymptotic standard error.

Using the preceding argument about the estimated variance, we can write

$$\text{se}(\hat{\beta}_j) \approx c_j/\sqrt{n}, \quad [5.10]$$

where c_j is a positive constant that does *not* depend on the sample size. In fact, the constant c_j can be shown to be

$$c_j = \frac{\sigma}{\sigma_j \sqrt{1 - \rho_j^2}}$$

where $\sigma = \text{sd}(u)$, $\sigma_j = \text{sd}(x_j)$, and ρ_j^2 is the population R -squared from regressing x_j on the other explanatory variables. Just like studying equation (5.9) to see which variables affect $\text{Var}(\hat{\beta}_j)$ under the Gauss-Markov assumptions, we can use this expression for c_j to study the impact of larger error standard deviation (σ), more population variation in x_j (σ_j), and multicollinearity in the population (ρ_j^2).

Equation (5.10) is only an approximation, but it is a useful rule of thumb: standard errors can be expected to shrink at a rate that is the inverse of the *square root* of the sample size.

EXAMPLE 5.2

STANDARD ERRORS IN A BIRTH WEIGHT EQUATION

We use the data in BWGHT.RAW to estimate a relationship where log of birth weight is the dependent variable, and cigarettes smoked per day (*cigs*) and log of family income are independent variables. The total number of observations is 1,388. Using the first half of the observations (694), the standard error for $\hat{\beta}_{cigs}$ is about .0013. The standard error using all of the observations is about .00086. The ratio of the latter standard error to the former is $.00086/.0013 \approx .662$. This is pretty close to $\sqrt{694/1,388} \approx .707$, the ratio obtained from the approximation in (5.10). In other words, equation (5.10) implies that the standard error using the larger sample size should be about 70.7% of the standard error using the smaller sample. This percentage is pretty close to the 66.2% we actually compute from the ratio of the standard errors.

The asymptotic normality of the OLS estimators also implies that the F statistics have approximate F distributions in large sample sizes. Thus, for testing exclusion restrictions or other multiple hypotheses, nothing changes from what we have done before.

Other Large Sample Tests: The Lagrange Multiplier Statistic

Once we enter the realm of asymptotic analysis, other test statistics can be used for hypothesis testing. For most purposes, there is little reason to go beyond the usual t and F statistics: as we just saw, these statistics have large sample justification without the normality assumption. Nevertheless, sometimes it is useful to have other ways to test multiple exclusion restrictions, and we now cover the **Lagrange multiplier (LM) statistic**, which has achieved some popularity in modern econometrics.

The name “Lagrange multiplier statistic” comes from constrained optimization, a topic beyond the scope of this text. [See Davidson and MacKinnon (1993).] The name **score statistic**—which also comes from optimization using calculus—is used as well. Fortunately, in the linear regression framework, it is simple to motivate the LM statistic without delving into complicated mathematics.

The form of the LM statistic we derive here relies on the Gauss-Markov assumptions, the same assumptions that justify the F statistic in large samples. We do not need the normality assumption.

To derive the LM statistic, consider the usual multiple regression model with k independent variables:

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u. \quad [5.11]$$

We would like to test whether, say, the last q of these variables all have zero population parameters: the null hypothesis is

$$H_0: \beta_{k-q+1} = 0, \dots, \beta_k = 0, \quad [5.12]$$

which puts q exclusion restrictions on the model (5.11). As with F testing, the alternative to (5.12) is that at least one of the parameters is different from zero.

The LM statistic requires estimation of the *restricted* model only. Thus, assume that we have run the regression

$$y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \dots + \tilde{\beta}_{k-q} x_{k-q} + \tilde{u}, \quad [5.13]$$

where “ $\sim$ ” indicates that the estimates are from the restricted model. In particular, $\tilde{u}$ indicates the residuals from the restricted model. (As always, this is just shorthand to indicate that we obtain the restricted residual for each observation in the sample.)

If the omitted variables x_{k-q+1} through x_k truly have zero population coefficients, then, at least approximately, $\tilde{u}$ should be uncorrelated with each of these variables in the sample. This suggests running a regression of these residuals on those independent variables excluded under H_0 , which is almost what the LM test does. However, it turns out that, to get a usable test statistic, we must include *all* of the independent variables in the regression. (We must include all regressors because, in general, the omitted regressors in the restricted model are correlated with the regressors that appear in the restricted model.) Thus, we run the regression of

$$\tilde{u} \text{ on } x_1, x_2, \dots, x_k. \quad [5.14]$$

This is an example of an **auxiliary regression**, a regression that is used to compute a test statistic but whose coefficients are not of direct interest.

How can we use the regression output from (5.14) to test (5.12)? If (5.12) is true, the R -squared from (5.14) should be “close” to zero, subject to sampling error, because $\tilde{u}$ will be approximately uncorrelated with all the independent variables. The question, as always with hypothesis testing, is how to determine when the statistic is large enough to reject the null hypothesis at a chosen significance level. It turns out that, under the null hypothesis, the sample size multiplied by the usual R -squared from the auxiliary regression (5.14) is distributed asymptotically as a chi-square random variable with q degrees of freedom. This leads to a simple procedure for testing the joint significance of a set of q independent variables.

The Lagrange Multiplier Statistic for q Exclusion Restrictions:

- (i) Regress y on the *restricted* set of independent variables and save the residuals, $\tilde{u}$.
- (ii) Regress $\tilde{u}$ on *all* of the independent variables and obtain the R -squared, say, R_u^2 (to distinguish it from the R -squareds obtained with y as the dependent variable).
- (iii) Compute $LM = nR_u^2$ [the sample size times the R -squared obtained from step (ii)].
- (iv) Compare LM to the appropriate critical value, c , in a χ_q^2 distribution; if $LM > c$, the null hypothesis is rejected. Even better, obtain the p -value as the probability that a χ_q^2 random variable exceeds the value of the test statistic. If the p -value is less than the desired significance level, then H_0 is rejected. If not, we fail to reject H_0 . The rejection rule is essentially the same as for F testing.

Because of its form, the LM statistic is sometimes referred to as the **n - R -squared statistic**. Unlike with the F statistic, the degrees of freedom in the unrestricted model plays no role in carrying out the LM test. All that matters is the number of restrictions being tested (q), the size of the auxiliary R -squared (R_u^2), and the sample size (n). The df in the unrestricted model plays no role because of the asymptotic nature of the LM statistic. But we must be sure to multiply R_u^2 by the sample size to obtain LM ; a seemingly low value of the R -squared can still lead to joint significance if n is large.

Before giving an example, a word of caution is in order. If in step (i), we mistakenly regress y on all of the independent variables and obtain the residuals from this unrestricted regression to be used in step (ii), we do not get an interesting statistic: the resulting R -squared will be exactly zero! This is because OLS chooses the estimates so that the residuals are uncorrelated in samples with all included independent variables [see equations in (3.13)]. Thus, we can only test (5.12) by regressing the restricted residuals on *all* of the independent variables. (Regressing the restricted residuals on the restricted set of independent variables will also produce $R^2 = 0$.)

EXAMPLE 5.3

ECONOMIC MODEL OF CRIME

We illustrate the LM test by using a slight extension of the crime model from Example 3.5:

$$narr86 = \beta_0 + \beta_1 pcnv + \beta_2 avg\ sen + \beta_3 tot\ time + \beta_4 pt\ ime86 + \beta_5 qemp86 + u,$$

where

$narr86$ = the number of times a man was arrested.

$pcnv$ = the proportion of prior arrests leading to conviction.

$avg\ sen$ = average sentence served from past convictions.

$tot\ time$ = total time the man has spent in prison prior to 1986 since reaching the age of 18.

$pt\ ime86$ = months spent in prison in 1986.

$qemp86$ = number of quarters in 1986 during which the man was legally employed.

We use the LM statistic to test the null hypothesis that $avg\ sen$ and $tot\ time$ have no effect on $narr86$ once the other factors have been controlled for.

In step (i), we estimate the restricted model by regressing $narr86$ on $pcnv$, $pt\ ime86$, and $qemp86$; the variables $avg\ sen$ and $tot\ time$ are excluded from this regression. We obtain the residuals $\tilde{u}$ from this regression, 2,725 of them. Next, we run the regression of

$$\tilde{u} \text{ on } pcnv, pt\ ime86, qemp86, avg\ sen, \text{ and } tot\ time; \quad [5.15]$$

as always, the order in which we list the independent variables is irrelevant. This second regression produces R_u^2 , which turns out to be about .0015. This may seem small, but we must multiply it by n to get the LM statistic: $LM = 2,725(.0015) \approx 4.09$. The 10% critical value in a chi-square distribution with two degrees of freedom is about 4.61 (rounded to two decimal places; see Table G.4). Thus, we fail to reject the null hypothesis that $\beta_{avg\ sen} = 0$ and $\beta_{tot\ time} = 0$ at the 10% level. The p -value is $P(\chi_2^2 > 4.09) \approx .129$, so we would reject H_0 at the 15% level.

As a comparison, the F test for joint significance of $avg\ sen$ and $tot\ time$ yields a p -value of about .131, which is pretty close to that obtained using the LM statistic. This is not surprising since, asymptotically, the two statistics have the same probability of Type I error. (That is, they reject the null hypothesis with the same frequency when the null is true.)

As the previous example suggests, with a large sample, we rarely see important discrepancies between the outcomes of LM and F tests. We will use the F statistic for the most part because it is computed routinely by most regression packages. But you should be aware of the LM statistic as it is used in applied work.

One final comment on the LM statistic. As with the F statistic, we must be sure to use the same observations in steps (i) and (ii). If data are missing for some of the independent variables that are excluded under the null hypothesis, the residuals from step (i) should be obtained from a regression on the reduced data set.

5.3 Asymptotic Efficiency of OLS

We know that, under the Gauss-Markov assumptions, the OLS estimators are best linear unbiased. OLS is also **asymptotically efficient** among a certain class of estimators under the Gauss-Markov assumptions. A general treatment requires matrix algebra and advanced asymptotic analysis. First, we describe the result in the simple regression case.

In the model

$$y = \beta_0 + \beta_1 x + u, \quad [5.16]$$

u has a zero conditional mean under MLR.4: $E(u|x) = 0$. This opens up a variety of consistent estimators for β_0 and β_1 ; as usual, we focus on the slope parameter, β_1 . Let $g(x)$ be any function of x ; for example, $g(x) = x^2$ or $g(x) = 1/(1 + |x|)$. Then u is uncorrelated with $g(x)$ (see Property CE.5 in Appendix B). Let $z_i = g(x_i)$ for all observations i . Then the estimator

$$\tilde{\beta}_1 = \left(\sum_{i=1}^n (z_i - \bar{z}) y_i \right) / \left(\sum_{i=1}^n (z_i - \bar{z}) x_i \right) \quad [5.17]$$

is consistent for β_1 , provided $g(x)$ and x are correlated. [Remember, it is possible that $g(x)$ and x are uncorrelated because correlation measures *linear* dependence.] To see this, we can plug in $y_i = \beta_0 + \beta_1 x_i + u_i$ and write $\tilde{\beta}_1$ as

$$\tilde{\beta}_1 = \beta_1 + \left(n^{-1} \sum_{i=1}^n (z_i - \bar{z}) u_i \right) / \left(n^{-1} \sum_{i=1}^n (z_i - \bar{z}) x_i \right). \quad [5.18]$$

Now, we can apply the law of large numbers to the numerator and denominator, which converge in probability to $\text{Cov}(z, u)$ and $\text{Cov}(z, x)$, respectively. Provided that $\text{Cov}(z, x) \neq 0$ —so that z and x are correlated—we have

$$\text{plim } \tilde{\beta}_1 = \beta_1 + \text{Cov}(z, u) / \text{Cov}(z, x) = \beta_1,$$

because $\text{Cov}(z, u) = 0$ under MLR.4.

It is more difficult to show that $\tilde{\beta}_1$ is asymptotically normal. Nevertheless, using arguments similar to those in the appendix, it can be shown that $\sqrt{n}(\tilde{\beta}_1 - \beta_1)$ is asymptotically normal with mean zero and asymptotic variance $\sigma^2 \text{Var}(z) / [\text{Cov}(z, x)]^2$. The asymptotic variance of the OLS estimator is obtained when $z = x$, in which case, $\text{Cov}(z, x) = \text{Cov}(x, x) = \text{Var}(x)$. Therefore, the asymptotic variance of $\sqrt{n}(\hat{\beta}_1 - \beta_1)$, where $\hat{\beta}_1$ is the OLS estimator, is $\sigma^2 \text{Var}(x) / [\text{Var}(x)]^2 = \sigma^2 / \text{Var}(x)$. Now, the Cauchy-Schwartz inequality (see Appendix B.4) implies that $[\text{Cov}(z, x)]^2 \leq \text{Var}(z) \text{Var}(x)$, which implies that the asymptotic variance of $\sqrt{n}(\tilde{\beta}_1 - \beta_1)$ is no larger than that of $\sqrt{n}(\hat{\beta}_1 - \beta_1)$. We have shown in the simple regression case that, under the Gauss-Markov assumptions, the OLS estimator has a smaller

asymptotic variance than any estimator of the form (5.17). [The estimator in (5.17) is an example of an *instrumental variables estimator*, which we will study extensively in Chapter 15.] If the homoskedasticity assumption fails, then there are estimators of the form (5.17) that have a smaller asymptotic variance than OLS. We will see this in Chapter 8.

The general case is similar but much more difficult mathematically. In the k regressor case, the class of consistent estimators is obtained by generalizing the OLS first order conditions:

$$\sum_{i=1}^n g_j(\mathbf{x}_i)(y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_{i1} - \dots - \tilde{\beta}_k x_{ik}) = 0, j = 0, 1, \dots, k, \quad [5.19]$$

where $g_j(\mathbf{x}_i)$ denotes any function of all explanatory variables for observation i . As can be seen by comparing (5.19) with the OLS first order conditions in (3.13), we obtain the OLS estimators when $g_0(\mathbf{x}_i) = 1$ and $g_j(\mathbf{x}_i) = x_{ij}$ for $j = 1, 2, \dots, k$. The class of estimators in (5.19) is infinite, because we can use any functions of the x_{ij} that we want.

THEOREM

5.3

ASYMPTOTIC EFFICIENCY OF OLS

Under the Gauss-Markov assumptions, let $\tilde{\beta}_j$ denote estimators that solve equations of the form (5.19) and let $\hat{\beta}_j$ denote the OLS estimators. Then for $j = 0, 1, 2, \dots, k$, the OLS estimators have the smallest asymptotic variances: $\text{Avar } \sqrt{n}(\hat{\beta}_j - \beta_j) \leq \text{Avar } \sqrt{n}(\tilde{\beta}_j - \beta_j)$.

Proving consistency of the estimators in (5.19), let alone showing they are asymptotically normal, is mathematically difficult. See Wooldridge (2010, Chapter 5).

Summary

The claims underlying the material in this chapter are fairly technical, but their practical implications are straightforward. We have shown that the first four Gauss-Markov assumptions imply that OLS is consistent. Furthermore, all of the methods of testing and constructing confidence intervals that we learned in Chapter 4 are approximately valid without assuming that the errors are drawn from a normal distribution (equivalently, the distribution of y given the explanatory variables is not normal). This means that we can apply OLS and use previous methods for an array of applications where the dependent variable is not even approximately normally distributed. We also showed that the *LM* statistic can be used instead of the *F* statistic for testing exclusion restrictions.

Before leaving this chapter, we should note that examples such as Example 5.3 may very well have problems that *do* require special attention. For a variable such as *narr86*, which is zero or one for most men in the population, a linear model may not be able to adequately capture the functional relationship between *narr86* and the explanatory variables. Moreover, even if a linear model does describe the expected value of arrests, heteroskedasticity might be a problem. Problems such as these are not mitigated as the sample size grows, and we will return to them in later chapters.

Key Terms

Asymptotic Bias	Asymptotic t Statistics	Lagrange Multiplier (LM) Statistic
Asymptotic Confidence Interval	Asymptotic Variance	Large Sample Properties
Asymptotic Normality	Asymptotically Efficient	n - R -Squared Statistic
Asymptotic Properties	Auxiliary Regression	Score Statistic
Asymptotic Standard Error	Consistency	
	Inconsistency	

Problems

- 1 In the simple regression model under MLR.1 through MLR.4, we argued that the slope estimator, $\hat{\beta}_1$, is consistent for β_1 . Using $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1$, show that $\text{plim } \hat{\beta}_0 = \beta_0$. [You need to use the consistency of $\hat{\beta}_1$ and the law of large numbers, along with the fact that $\beta_0 = E(y) - \beta_1 E(x_1)$.]
- 2 Suppose that the model

$$pctstck = \beta_0 + \beta_1 funds + \beta_2 risktol + u$$

satisfies the first four Gauss-Markov assumptions, where *pctstck* is the percentage of a worker's pension invested in the stock market, *funds* is the number of mutual funds that the worker can choose from, and *risktol* is some measure of risk tolerance (larger *risktol* means the person has a higher tolerance for risk). If *funds* and *risktol* are positively correlated, what is the inconsistency in $\hat{\beta}_1$, the slope coefficient in the simple regression of *pctstck* on *funds*?

- 3 The data set SMOKE.RAW contains information on smoking behavior and other variables for a random sample of single adults from the United States. The variable *cigs* is the (average) number of cigarettes smoked per day. Do you think *cigs* has a normal distribution in the U.S. adult population? Explain.
- 4 In the simple regression model (5.16), under the first four Gauss-Markov assumptions, we showed that estimators of the form (5.17) are consistent for the slope, β_1 . Given such an estimator, define an estimator of β_0 by $\tilde{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Show that $\text{plim } \tilde{\beta}_0 = \beta_0$.

Computer Exercises

- C1 Use the data in WAGE1.RAW for this exercise.
 - (i) Estimate the equation

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u.$$

Save the residuals and plot a histogram.

- (ii) Repeat part (i), but with $\log(wage)$ as the dependent variable.
- (iii) Would you say that Assumption MLR.6 is closer to being satisfied for the level-level model or the log-level model?

C2 Use the data in GPA2.RAW for this exercise.

- (i) Using all 4,137 observations, estimate the equation

$$\text{colgpa} = \beta_0 + \beta_1 \text{hsperc} + \beta_2 \text{sat} + u$$

and report the results in standard form.

- (ii) Reestimate the equation in part (i), using the first 2,070 observations.
 (iii) Find the ratio of the standard errors on *hsperc* from parts (i) and (ii). Compare this with the result from (5.10).

C3 In equation (4.42) of Chapter 4, using the data set BWGHT.RAW, compute the *LM* statistic for testing whether *motheduc* and *fatheduc* are jointly significant. In obtaining the residuals for the restricted model, be sure that the restricted model is estimated using only those observations for which all variables in the unrestricted model are available (see Example 4.9).

C4 Several statistics are commonly used to detect nonnormality in underlying population distributions. Here we will study one that measures the amount of skewness in a distribution. Recall that any normally distributed random variable is symmetric about its mean; therefore, if we standardize a symmetrically distributed random variable, say $z = (y - \mu_y)/\sigma_y$, where $\mu_y = E(y)$ and $\sigma_y = \text{sd}(y)$, then z has mean zero, variance one, and $E(z^3) = 0$. Given a sample of data $\{y_i : i = 1, \dots, n\}$, we can standardize y_i in the sample by using $z_i = (y_i - \hat{\mu}_y)/\hat{\sigma}_y$, where $\hat{\mu}_y$ is the sample mean and $\hat{\sigma}_y$ is the sample standard deviation. (We ignore the fact that these are estimates based on the sample.) A sample statistic that measures skewness is $n^{-1} \sum_{i=1}^n z_i^3$, or where n is replaced with $(n - 1)$ as a degrees-of-freedom adjustment. If y has a normal distribution in the population, the skewness measure in the sample for the standardized values should not differ significantly from zero.

- (i) First use the data set 401KSUBS.RAW, keeping only observations with *fsize* = 1. Find the skewness measure for *inc*. Do the same for $\log(\text{inc})$. Which variable has more skewness and therefore seems less likely to be normally distributed?
 (ii) Next use BWGHT2.RAW. Find the skewness measures for *bwght* and $\log(\text{bwght})$. What do you conclude?
 (iii) Evaluate the following statement: “The logarithmic transformation always makes a positive variable look more normally distributed.”
 (iv) If we are interested in the normality assumption in the context of regression, should we be evaluating the unconditional distributions of y and $\log(y)$? Explain.

C5 Consider the analysis in Computer Exercise C11 in Chapter 4 using the data in HTV.RAW, where *educ* is the dependent variable in a regression.

- (i) How many different values are taken on by *educ* in the sample? Does *educ* have a continuous distribution?
 (ii) Plot a histogram of *educ* with a normal distribution overlay. Does the distribution of *educ* appear anything close to normal?
 (iii) Which of the CLM assumptions seems clearly violated in the model

$$\text{educ} = \beta_0 + \beta_1 \text{motheduc} + \beta_2 \text{fatheduc} + \beta_3 \text{abil} + \beta_4 \text{abil}^2 + u?$$

How does this violation change the statistical inference procedures carried out in Computer Exercise C11 in Chapter 4?

APPENDIX 5A

Asymptotic Normality of OLS

We sketch a proof of the asymptotic normality of OLS [Theorem 5.2(i)] in the simple regression case. Write the simple regression model as in equation (5.16). Then, by the usual algebra of simple regression, we can write

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = (1/s_x^2) \left[n^{-1/2} \sum_{i=1}^n (x_i - \bar{x}) u_i \right],$$

where we use s_x^2 to denote the sample variance of $\{x_i; i = 1, 2, \dots, n\}$. By the law of large numbers (see Appendix C), $s_x^2 \xrightarrow{p} \sigma_x^2 = \text{Var}(x)$. Assumption MLR.3 rules out no perfect collinearity, which means that $\text{Var}(x) > 0$ (x_i varies in the sample, and therefore x is not constant in the population). Next, $n^{-1/2} \sum_{i=1}^n (x_i - \bar{x}) u_i = n^{-1/2} \sum_{i=1}^n (x_i - \mu) u_i + (\mu - \bar{x}) [n^{-1/2} \sum_{i=1}^n u_i]$, where $\mu = E(x)$ is the population mean of x . Now $\{u_i\}$ is a sequence of i.i.d. random variables with mean zero and variance σ^2 , and so $n^{-1/2} \sum_{i=1}^n u_i$ converges to the $\text{Normal}(0, \sigma^2)$ distribution as $n \rightarrow \infty$; this is just the central limit theorem from Appendix C. By the law of large numbers, $\text{plim}(\mu - \bar{x}) = 0$. A standard result in asymptotic theory is that if $\text{plim}(w_n) = 0$ and z_n has an asymptotic normal distribution, then $\text{plim}(w_n z_n) = 0$. [See Wooldridge (2010, Chapter 3) for more discussion.] This implies that $(\mu - \bar{x}) [n^{-1/2} \sum_{i=1}^n u_i]$ has zero plim. Next, $\{(x_i - \mu) u_i; i = 1, 2, \dots\}$ is an indefinite sequence of i.i.d. random variables with mean zero—because u and x are uncorrelated under Assumption MLR.4—and variance $\sigma^2 \sigma_x^2$ by the homoskedasticity Assumption MLR.5. Therefore, $n^{-1/2} \sum_{i=1}^n (x_i - \mu) u_i$ has an asymptotic $\text{Normal}(0, \sigma^2 \sigma_x^2)$ distribution. We just showed that the difference between $n^{-1/2} \sum_{i=1}^n (x_i - \bar{x}) u_i$ and $n^{-1/2} \sum_{i=1}^n (x_i - \mu) u_i$ has zero plim. A result in asymptotic theory is that if z_n has an asymptotic normal distribution and $\text{plim}(v_n - z_n) = 0$, then v_n has the same asymptotic normal distribution as z_n . It follows that $n^{-1/2} \sum_{i=1}^n (x_i - \bar{x}) u_i$ also has an asymptotic $\text{Normal}(0, \sigma^2 \sigma_x^2)$ distribution. Putting all of the pieces together gives

$$\begin{aligned} \sqrt{n}(\hat{\beta}_1 - \beta_1) &= (1/\sigma_x^2) \left[n^{-1/2} \sum_{i=1}^n (x_i - \bar{x}) u_i \right] \\ &\quad + [(1/s_x^2) - (1/\sigma_x^2)] \left[n^{-1/2} \sum_{i=1}^n (x_i - \bar{x}) u_i \right], \end{aligned}$$

and since $\text{plim}(1/s_x^2) = 1/\sigma_x^2$, the second term has zero plim. Therefore, the asymptotic distribution of $\sqrt{n}(\hat{\beta}_1 - \beta_1)$ is $\text{Normal}(0, \{\sigma^2 \sigma_x^2\} / \{\sigma_x^2\}^2) = \text{Normal}(0, \sigma^2 / \sigma_x^2)$. This completes the proof in the simple regression case, as $a_1^2 = \sigma_x^2$ in this case. See Wooldridge (2010, Chapter 4) for the general case.

Multiple Regression Analysis: Further Issues

This chapter brings together several issues in multiple regression analysis that we could not conveniently cover in earlier chapters. These topics are not as fundamental as the material in Chapters 3 and 4, but they are important for applying multiple regression to a broad range of empirical problems.

6.1 Effects of Data Scaling on OLS Statistics

In Chapter 2 on bivariate regression, we briefly discussed the effects of changing the units of measurement on the OLS intercept and slope estimates. We also showed that changing the units of measurement did not affect R -squared. We now return to the issue of data scaling and examine the effects of rescaling the dependent or independent variables on standard errors, t statistics, F statistics, and confidence intervals.

We will discover that everything we expect to happen, does happen. When variables are rescaled, the coefficients, standard errors, confidence intervals, t statistics, and F statistics change in ways that preserve all measured effects and testing outcomes. Although this is no great surprise—in fact, we would be very worried if it were not the case—it is useful to see what occurs explicitly. Often, data scaling is used for cosmetic purposes, such as to reduce the number of zeros after a decimal point in an estimated coefficient. By judiciously choosing units of measurement, we can improve the appearance of an estimated equation while changing nothing that is essential.

We could treat this problem in a general way, but it is much better illustrated with examples. Likewise, there is little value here in introducing an abstract notation.

We begin with an equation relating infant birth weight to cigarette smoking and family income:

$$\widehat{bwght} = \hat{\beta}_0 + \hat{\beta}_1cigs + \hat{\beta}_2faminc, \quad [6.1]$$

where

$bwght$ = child birth weight, in ounces.

$cigs$ = number of cigarettes smoked by the mother while pregnant, per day.

$faminc$ = annual family income, in thousands of dollars.

TABLE 6.1 Effects of Data Scaling

Dependent Variable	(1) <i>bwght</i>	(2) <i>bwghtlbs</i>	(3) <i>bwght</i>
Independent Variables			
<i>cigs</i>	-.4634 (.0916)	-.0289 (.0057)	—
<i>packs</i>	—	—	-9.268 (1.832)
<i>faminc</i>	.0927 (.0292)	.0058 (.0018)	.0927 (.0292)
<i>intercept</i>	116.974 (1.049)	7.3109 (.0656)	116.974 (1.049)
Observations	1,388	1,388	1,388
R-Squared	.0298	.0298	.0298
SSR	557,485.51	2,177.6778	557,485.51
SER	20.063	1.2539	20.063

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The estimates of this equation, obtained using the data in BWGHT.RAW, are given in the first column of Table 6.1. Standard errors are listed in parentheses. The estimate on *cigs* says that if a woman smoked 5 more cigarettes per day, birth weight is predicted to be about $.4634(5) = 2.317$ ounces less. The *t* statistic on *cigs* is -5.06 , so the variable is very statistically significant.

Now, suppose that we decide to measure birth weight in pounds, rather than in ounces. Let $bwghtlbs = bwght/16$ be birth weight in pounds. What happens to our OLS statistics if we use this as the dependent variable in our equation? It is easy to find the effect on the coefficient estimates by simple manipulation of equation (6.1). Divide this entire equation by 16:

$$\widehat{bwght}/16 = \hat{\beta}_0/16 + (\hat{\beta}_1/16)cigs + (\hat{\beta}_2/16)faminc.$$

Since the left-hand side is birth weight in pounds, it follows that each new coefficient will be the corresponding old coefficient divided by 16. To verify this, the regression of *bwghtlbs* on *cigs*, and *faminc* is reported in column (2) of Table 6.1. Up to four digits, the intercept and slopes in column (2) are just those in column (1) divided by 16. For example, the coefficient on *cigs* is now $-.0289$; this means that if *cigs* were higher by five, birth weight would be $.0289(5) = .1445$ pounds lower. In terms of ounces, we have $.1445(16) = 2.312$, which is slightly different from the 2.317 we obtained earlier due to rounding error. The point is, once the effects are transformed into the same units, we get exactly the same answer, regardless of how the dependent variable is measured.

What about statistical significance? As we expect, changing the dependent variable from ounces to pounds has no effect on how statistically important the independent variables are. The standard errors in column (2) are 16 times smaller than those in column (1). A few quick calculations show that the *t* statistics in column (2) are indeed identical to the *t* statistics in column (1). The endpoints for the confidence intervals in column (2) are just the endpoints in column (1) divided by 16. This is because the CIs change by the same factor as the standard errors. [Remember that the 95% CI here is $\hat{\beta}_j \pm 1.96 \text{ se}(\hat{\beta}_j)$.]

In terms of goodness-of-fit, the R -squareds from the two regressions are identical, as should be the case. Notice that the sum of squared residuals, SSR, and the standard error of the regression, SER, do differ across equations. These differences are easily explained. Let $\hat{u}_i$ denote the residual for observation i in the original equation (6.1). Then the residual when *bwghtlbs* is the dependent variable is simply $\hat{u}_i/16$. Thus, the *squared* residual in the second equation is $(\hat{u}_i/16)^2 = \hat{u}_i^2/256$. This is why the sum of squared residuals in column (2) is equal to the SSR in column (1) divided by 256.

Since $SER = \hat{\sigma} = \sqrt{SSR/(n - k - 1)} = \sqrt{SSR/1,385}$, the SER in column (2) is 16 times smaller than that in column (1). Another way to think about this is that the error in the equation with *bwghtlbs* as the dependent variable has a standard deviation 16 times smaller than the standard deviation of the original error. This does not mean that we have reduced the error by changing how birth weight is measured; the smaller SER simply reflects a difference in units of measurement.

Next, let us return the dependent variable to its original units: *bwght* is measured in ounces. Instead, let us change the unit of measurement of one of the independent variables, *cigs*. Define *packs* to be the number of packs of cigarettes smoked per day. Thus, *packs* = *cigs*/20. What happens to the coefficients and other OLS statistics now? Well, we can write

$$\widehat{bwght} = \hat{\beta}_0 + (20\hat{\beta}_1)(cigs/20) + \hat{\beta}_2 faminc = \hat{\beta}_0 + (20\hat{\beta}_1)packs + \hat{\beta}_2 faminc.$$

Thus, the intercept and slope coefficient on *faminc* are unchanged, but the coefficient on *packs* is 20 times that on *cigs*. This is intuitively appealing. The results from the regression of *bwght* on *packs* and *faminc* are in column (3) of Table 6.1. Incidentally, remember that it would make no sense to include both *cigs* and *packs* in the same equation; this would induce perfect multicollinearity and would have no interesting meaning.

Other than the coefficient on *packs*, there is one other statistic in column (3) that differs from that in column (1): the standard error on *packs* is 20 times larger than that on *cigs* in column (1). This means that the t statistic for testing the significance of cigarette smoking is the same whether we measure smoking in terms of cigarettes or packs. This is only natural.

EXPLORING FURTHER 6.1

In the original birth weight equation (6.1), suppose that *faminc* is measured in dollars rather than in thousands of dollars. Thus, define the variable *fincdol* = 1,000·*faminc*. How will the OLS statistics change when *fincdol* is substituted for *faminc*? For the purpose of presenting the regression results, do you think it is better to measure income in dollars or in thousands of dollars?

The previous example spells out most of the possibilities that arise when the dependent and independent variables are rescaled. Rescaling is often done with dollar amounts in economics, especially when the dollar amounts are very large.

In Chapter 2, we argued that, if the dependent variable appears in logarithmic form, changing the unit of measurement does not affect the slope coefficient. The same is true here: changing the unit of measurement of the dependent variable, when it appears in logarithmic form, does not affect any of the slope estimates. This follows from the simple fact that $\log(c_1 y_i) = \log(c_1) + \log(y_i)$ for any constant $c_1 > 0$. The new intercept will be $\log(c_1) + \hat{\beta}_0$. Similarly, changing the unit of measurement of any x_j , where $\log(x_j)$ appears in the regression, only affects the intercept. This corresponds to what we know about percentage changes and, in particular, elasticities: they are invariant to the units of measurement of either y or the x_j . For example, if we had specified the dependent

variable in (6.1) to be $\log(bwght)$, estimated the equation, and then reestimated it with $\log(bwghtlbs)$ as the dependent variable, the coefficients on *cigs* and *faminc* would be the same in both regressions; only the intercept would be different.

Beta Coefficients

Sometimes, in econometric applications, a key variable is measured on a scale that is difficult to interpret. Labor economists often include test scores in wage equations, and the scale on which these tests are scored is often arbitrary and not easy to interpret (at least for economists!). In almost all cases, we are interested in how a particular individual's score compares with the population. Thus, instead of asking about the effect on hourly wage if, say, a test score is 10 points higher, it makes more sense to ask what happens when the test score is one *standard deviation* higher.

Nothing prevents us from seeing what happens to the dependent variable when an independent variable in an estimated model increases by a certain number of standard deviations, assuming that we have obtained the sample standard deviation (which is easy in most regression packages). This is often a good idea. So, for example, when we look at the effect of a standardized test score, such as the SAT score, on college GPA, we can find the standard deviation of SAT and see what happens when the SAT score increases by one or two standard deviations.

Sometimes, it is useful to obtain regression results when *all* variables involved, the dependent as well as all the independent variables, have been *standardized*. A variable is standardized in the sample by subtracting off its mean and dividing by its standard deviation (see Appendix C). This means that we compute the *z-score* for every variable in the sample. Then, we run a regression using the *z-scores*.

Why is standardization useful? It is easiest to start with the original OLS equation, with the variables in their original forms:

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_k x_{ik} + \hat{u}_i. \quad [6.2]$$

We have included the observation subscript *i* to emphasize that our standardization is applied to all sample values. Now, if we average (6.2), use the fact that the $\hat{u}_i$ have a zero sample average, and subtract the result from (6.2), we get

$$y_i - \bar{y} = \hat{\beta}_1 (x_{i1} - \bar{x}_1) + \hat{\beta}_2 (x_{i2} - \bar{x}_2) + \dots + \hat{\beta}_k (x_{ik} - \bar{x}_k) + \hat{u}_i.$$

Now, let $\hat{\sigma}_y$ be the sample standard deviation for the dependent variable, let $\hat{\sigma}_1$ be the sample *sd* for x_1 , let $\hat{\sigma}_2$ be the sample *sd* for x_2 , and so on. Then, simple algebra gives the equation

$$\begin{aligned} (y_i - \bar{y})/\hat{\sigma}_y &= (\hat{\sigma}_1/\hat{\sigma}_y)\hat{\beta}_1[(x_{i1} - \bar{x}_1)/\hat{\sigma}_1] + \dots \\ &+ (\hat{\sigma}_k/\hat{\sigma}_y)\hat{\beta}_k[(x_{ik} - \bar{x}_k)/\hat{\sigma}_k] + (\hat{u}_i/\hat{\sigma}_y). \end{aligned} \quad [6.3]$$

Each variable in (6.3) has been standardized by replacing it with its *z-score*, and this has resulted in **new slope coefficients**. For example, the slope coefficient on $(x_{i1} - \bar{x}_1)/\hat{\sigma}_1$ is $(\hat{\sigma}_1/\hat{\sigma}_y)\hat{\beta}_1$. This is simply the original coefficient, $\hat{\beta}_1$, multiplied by the ratio of the standard deviation of x_1 to the standard deviation of *y*. The intercept has dropped out altogether.

It is useful to rewrite (6.3), dropping the *i* subscript, as

$$z_y = \hat{b}_1 z_1 + \hat{b}_2 z_2 + \dots + \hat{b}_k z_k + \text{error}, \quad [6.4]$$

where z_y denotes the z -score of y , z_1 is the z -score of x_1 , and so on. The new coefficients are

$$\hat{b}_j = (\hat{\sigma}_j / \hat{\sigma}_y) \hat{\beta}_j \text{ for } j = 1, \dots, k. \quad [6.5]$$

These $\hat{b}_j$ are traditionally called **standardized coefficients** or **beta coefficients**. (The latter name is more common, which is unfortunate because we have been using beta hat to denote the usual OLS estimates.)

Beta coefficients receive their interesting meaning from equation (6.4): If x_1 increases by one standard deviation, then $\hat{y}$ changes by $\hat{b}_1$ standard deviations. Thus, we are measuring effects not in terms of the original units of y or the x_j , but in standard deviation units. Because it makes the scale of the regressors irrelevant, this equation puts the explanatory variables on equal footing. In a standard OLS equation, it is not possible to simply look at the size of different coefficients and conclude that the explanatory variable with the largest coefficient is “the most important.” We just saw that the magnitudes of coefficients can be changed at will by changing the units of measurement of the x_j . But, when each x_j has been standardized, comparing the magnitudes of the resulting beta coefficients is more compelling. When the regression equation has only a single explanatory variable, x_1 , its standardized coefficient is simply the sample correlation coefficient between y and x_1 , which means it must lie in the range -1 to 1 .

Even in situations where the coefficients are easily interpretable—say, the dependent variable and independent variables of interest are in logarithmic form, so the OLS coefficients of interest are estimated elasticities—there is still room for computing beta coefficients. Although elasticities are free of units of measurement, a change in a particular explanatory variable by, say, 10% may represent a larger or smaller change over a variable’s range than changing another explanatory variable by 10%. For example, in a state with wide income variation but relatively little variation in spending per student, it might not make much sense to compare performance elasticities with respect to the income and spending. Comparing beta coefficient magnitudes can be helpful.

To obtain the beta coefficients, we can always standardize y , x_1 , ..., x_k and then run the OLS regression of the z -score of y on the z -scores of x_1 , ..., x_k —where it is not necessary to include an intercept, as it will be zero. This can be tedious with many independent variables. Some regression packages provide beta coefficients via a simple command. The following example illustrates the use of beta coefficients.

EXAMPLE 6.1

EFFECTS OF POLLUTION ON HOUSING PRICES

We use the data from Example 4.5 (in the file HPRICE2.RAW) to illustrate the use of beta coefficients. Recall that the key independent variable is *nox*, a measure of the nitrogen oxide in the air over each community. One way to understand the size of the pollution effect—without getting into the science underlying nitrogen oxide’s effect on air quality—is to compute beta coefficients. (An alternative approach is contained in Example 4.5: we obtained a price elasticity with respect to *nox* by using *price* and *nox* in logarithmic form.)

The population equation is the level-level model

$$price = \beta_0 + \beta_1 nox + \beta_2 crime + \beta_3 rooms + \beta_4 dist + \beta_5 stratio + u,$$

where all the variables except *crime* were defined in Example 4.5; *crime* is the number of reported crimes per capita. The beta coefficients are reported in the following equation (so each variable has been converted to its *z*-score):

$$\widehat{zprice} = -.340 znox - .143 zcrime + .514 zrooms - .235 zdist - .270 zstratio.$$

This equation shows that a one standard deviation increase in *nox* decreases price by .34 standard deviation; a one standard deviation increase in *crime* reduces price by .14 standard deviation. Thus, the same relative movement of pollution in the population has a larger effect on housing prices than crime does. Size of the house, as measured by number of rooms (*rooms*), has the largest standardized effect. **If we want to know the effects of each independent variable on the dollar value of median house price, we should use the unstandardized variables.**

Whether we use standardized or unstandardized variables does not affect statistical significance: the *t* statistics are the same in both cases.

6.2 More on Functional Form

In several previous examples, we have encountered the most popular device in econometrics for allowing nonlinear relationships between the explained and explanatory variables: using logarithms for the dependent or independent variables. We have also seen models containing quadratics in some explanatory variables, but we have yet to provide a systematic treatment of them. In this section, we cover some variations and extensions on functional forms that often arise in applied work.

More on Using Logarithmic Functional Forms

We begin by reviewing how to interpret the parameters in the model

$$\log(\text{price}) = \beta_0 + \beta_1 \log(\text{nox}) + \beta_2 \text{rooms} + u, \quad [6.6]$$

where these variables are taken from Example 4.5. Recall that throughout the text $\log(x)$ is the *natural* log of x . The coefficient β_1 is the elasticity of *price* with respect to *nox* (pollution). The coefficient β_2 is the change in $\log(\text{price})$, when $\Delta \text{rooms} = 1$; as we have seen many times, when multiplied by 100, this is the approximate percentage change in *price*. Recall that $100 \cdot \beta_2$ is sometimes called the semi-elasticity of *price* with respect to *rooms*.

When estimated using the data in HPRICE2.RAW, we obtain

$$\begin{aligned} \widehat{\log(\text{price})} &= 9.23 - .718 \log(\text{nox}) + .306 \text{rooms} \\ &\quad (0.19) \quad (.066) \quad (.019) \\ n &= 506, R^2 = .514. \end{aligned} \quad [6.7]$$

Thus, when *nox* increases by 1%, *price* falls by .718%, holding only *rooms* fixed. When *rooms* increases by one, *price* increases by approximately $100(.306) = 30.6\%$.

The estimate that one more room increases price by about 30.6% turns out to be somewhat inaccurate for this application. The approximation error occurs because, as the change in $\log(y)$ becomes larger and larger, the approximation $\% \Delta y \approx 100 \cdot \Delta \log(y)$

becomes more and more inaccurate. Fortunately, a simple calculation is available to compute the exact percentage change.

To describe the procedure, we consider the general estimated model

$$\widehat{\log(y)} = \hat{\beta}_0 + \hat{\beta}_1 \log(x_1) + \hat{\beta}_2 x_2.$$

(Adding additional independent variables does not change the procedure.) Now, fixing x_1 , we have $\Delta \widehat{\log(y)} = \hat{\beta}_2 \Delta x_2$. Using simple algebraic properties of the exponential and logarithmic functions gives the *exact* percentage change in the predicted y as

$$\% \Delta \hat{y} = 100 \cdot [\exp(\hat{\beta}_2 \Delta x_2) - 1], \quad [6.8]$$

where the multiplication by 100 turns the proportionate change into a percentage change. When $\Delta x_2 = 1$,

$$\% \Delta \hat{y} = 100 \cdot [\exp(\hat{\beta}_2) - 1]. \quad [6.9]$$

Applied to the housing price example with $x_2 = \text{rooms}$ and $\hat{\beta}_2 = .306$, $\% \Delta \widehat{price} = 100[\exp(.306) - 1] = 35.8\%$, which is notably larger than the approximate percentage change, 30.6%, obtained directly from (6.7). {Incidentally, this is not an unbiased estimator because $\exp(\cdot)$ is a nonlinear function; it is, however, a consistent estimator of $100[\exp(\beta_2) - 1]$. This is because the probability limit passes through continuous functions, while the expected value operator does not. See Appendix C.}

The adjustment in equation (6.8) is not as crucial for small percentage changes. For example, when we include the student-teacher ratio in equation (6.7), its estimated coefficient is $-.052$, which means that if *stratio* increases by one, *price* decreases by approximately 5.2%. The exact proportionate change is $\exp(-.052) - 1 \approx -.051$, or -5.1% . On the other hand, if we increase *stratio* by five, then the approximate percentage change in price is -26% , while the exact change obtained from equation (6.8) is $100[\exp(-.26) - 1] \approx -22.9\%$.

The logarithmic approximation to percentage changes has an advantage that justifies its reporting even when the percentage change is large. To describe this advantage, consider again the effect on price of changing the number of rooms by one. The logarithmic approximation is just the coefficient on rooms in equation (6.7) multiplied by 100, namely, 30.6%. We also computed an estimate of the exact percentage change for *increasing* the number of rooms by one as 35.8%. But what if we want to estimate the percentage change for *decreasing* the number of rooms by one? In equation (6.8) we take $\Delta x_2 = -1$ and $\hat{\beta}_2 = .306$, and so $\% \Delta \widehat{price} = 100[\exp(-.306) - 1] = -26.4$, or a drop of 26.4%. Notice that the approximation based on using the coefficient on *rooms* is between 26.4 and 35.8—an outcome that always occurs. In other words, simply using the coefficient (multiplied by 100) gives us an estimate that is always between the absolute value of the estimates for an increase and a decrease. If we are specifically interested in an increase or a decrease, we can use the calculation based on equation (6.8).

The point just made about computing percentage changes is essentially the one made in introductory economics when it comes to computing, say, price elasticities of demand based on large price changes: the result depends on whether we use the beginning or ending price and quantity in computing the percentage changes. Using the logarithmic approximation is similar in spirit to calculating an arc elasticity of demand, where the average of prices and quantities is used in the denominators in computing the percentage changes.

We have seen that using natural logs leads to coefficients with appealing interpretations, and we can be ignorant about the units of measurement of variables appearing in logarithmic form because the slope coefficients are invariant to rescalings. There are several

other reasons logs are used so much in applied work. First, when $y > 0$, models using $\log(y)$ as the dependent variable often satisfy the CLM assumptions more closely than models using the level of y . Strictly positive variables often have conditional distributions that are heteroskedastic or skewed; taking the log can mitigate, if not eliminate, both problems.

Another potential benefit of using logs is that taking the log of a variable often narrows its range. This is particularly true of variables that can be large monetary values, such as firms' annual sales or baseball players' salaries. Population variables also tend to vary widely. Narrowing the range of the dependent and independent variables can make OLS estimates less sensitive to outlying (or extreme values); we take up the issue of outlying observations in Chapter 9.

However, one must not indiscriminantly use the logarithmic transformation because in some cases it can actually create extreme values. An example is when a variable y is between zero and one (such as a proportion) and takes on values close to zero. In this case, $\log(y)$ (which is necessarily negative) can be very large in magnitude whereas the original variable, y , is bounded between zero and one.

There are some standard rules of thumb for taking logs, although none is written in stone. When a variable is a positive dollar amount, the log is often taken. We have seen this for variables such as wages, salaries, firm sales, and firm market value. Variables such as population, total number of employees, and school enrollment often appear in logarithmic form; these have the common feature of being large integer values.

Variables that are measured in years—such as education, experience, tenure, age, and so on—usually appear in their original form. A variable that is a proportion or a percent—such as the unemployment rate, the participation rate in a pension plan, the percentage of students passing a standardized exam, and the arrest rate on reported crimes—can appear in either original or logarithmic form, although there is a tendency to use them in level forms. This is because any regression coefficients involving the *original* variable—whether it is the dependent or independent variable—will have a *percentage point* change interpretation. (See Appendix A for a review of the distinction between a percentage change and a percentage point change.) If we use, say, $\log(unem)$ in a regression, where $unem$ is the percentage of unemployed individuals, we must be very careful to distinguish between a percentage point change and a percentage change. Remember, if $unem$ goes from 8 to 9, this is an increase of one percentage point, but a 12.5% increase from the initial unemployment level. Using the log means that we are looking at the percentage change in the unemployment rate: $\log(9) - \log(8) \approx .118$ or 11.8%, which is the logarithmic approximation to the actual 12.5% increase.

EXPLORING FURTHER 6.2

Suppose that the annual number of drunk driving arrests is determined by

$$\log(arrests) = \beta_0 + \beta_1 \log(pop) + \beta_2 age_{16_25} + \text{other factors},$$

where age_{16_25} is the proportion of the population between 16 and 25 years of age. Show that β_2 has the following (ceteris paribus) interpretation: it is the percentage change in *arrests* when the percentage of the people aged 16 to 25 increases by one *percentage point*.

One limitation of the log is that it cannot be used if a variable takes on zero or negative values. In cases where a variable y is nonnegative but can take on the value 0, $\log(1+y)$ is sometimes used. The percentage change interpretations are often closely preserved, except for changes beginning at $y = 0$ (where the percentage change is not even defined). Generally, using $\log(1+y)$ and then interpreting the estimates as if the variable were $\log(y)$ is acceptable when the data on y contain relatively few zeros. An example might be where y is hours of training per employee for the population of manufacturing firms,

if a large fraction of firms provides training to at least one worker. Technically, however, $\log(1 + y)$ cannot be normally distributed (although it might be less heteroskedastic than y). Useful, albeit more advanced, alternatives are the Tobit and Poisson models in Chapter 17.

One drawback to using a dependent variable in logarithmic form is that it is more difficult to predict the original variable. The original model allows us to predict $\log(y)$, not y . Nevertheless, it is fairly easy to turn a prediction for $\log(y)$ into a prediction for y (see Section 6.4). A related point is that it is *not* legitimate to compare R -squareds from models where y is the dependent variable in one case and $\log(y)$ is the dependent variable in the other. These measures explain variations in different variables. We discuss how to compute comparable goodness-of-fit measures in Section 6.4.

Models with Quadratics

Quadratic functions are also used quite often in applied economics to capture decreasing or increasing marginal effects. You may want to review properties of quadratic functions in Appendix A.

In the simplest case, y depends on a single observed factor x , but it does so in a quadratic fashion:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u.$$

For example, take $y = \text{wage}$ and $x = \text{exper}$. As we discussed in Chapter 3, this model falls outside of simple regression analysis but is easily handled with multiple regression.

It is important to remember that β_1 does not measure the change in y with respect to x ; it makes no sense to hold x^2 fixed while changing x . If we write the estimated equation as

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2, \quad [6.10]$$

then we have the approximation

$$\Delta \hat{y} \approx (\hat{\beta}_1 + 2\hat{\beta}_2 x) \Delta x, \text{ so } \Delta \hat{y} / \Delta x \approx \hat{\beta}_1 + 2\hat{\beta}_2 x. \quad [6.11]$$

This says that the slope of the relationship between x and y depends on the value of x ; the estimated slope is $\hat{\beta}_1 + 2\hat{\beta}_2 x$. If we plug in $x = 0$, we see that $\hat{\beta}_1$ can be interpreted as the approximate slope in going from $x = 0$ to $x = 1$. After that, the second term, $2\hat{\beta}_2 x$, must be accounted for.

If we are only interested in computing the predicted change in y given a starting value for x and a change in x , we could use (6.10) directly: there is no reason to use the calculus approximation at all. However, we are usually more interested in quickly summarizing the effect of x on y , and the interpretation of $\hat{\beta}_1$ and $\hat{\beta}_2$ in equation (6.11) provides that summary. Typically, we might plug in the average value of x in the sample, or some other interesting values, such as the median or the lower and upper quartile values.

In many applications, $\hat{\beta}_1$ is positive and $\hat{\beta}_2$ is negative. For example, using the wage data in WAGE1.RAW, we obtain

$$\begin{aligned} \widehat{\text{wage}} &= 3.73 + .298 \text{ exper} - .0061 \text{ exper}^2 \\ &\quad (.35) \quad (.041) \quad \quad (.0009) \\ n &= 526, R^2 = .093. \end{aligned} \quad [6.12]$$

This estimated equation implies that *exper* has a diminishing effect on *wage*. The first year of experience is worth roughly 30¢ per hour (\$.298). The second year of experience is worth less [about $.298 - 2(.0061)(1) \approx .286$, or 28.6¢, according to the approximation in (6.11) with $x = 1$]. In going from 10 to 11 years of experience, *wage* is predicted to increase by about $.298 - 2(.0061)(10) = .176$, or 17.6¢. And so on.

When the coefficient on x is positive and the coefficient on x^2 is negative, the quadratic has a parabolic shape. There is always a positive value of x where the effect of x on y is zero; before this point, x has a positive effect on y ; after this point, x has a negative effect on y . In practice, it can be important to know where this turning point is.

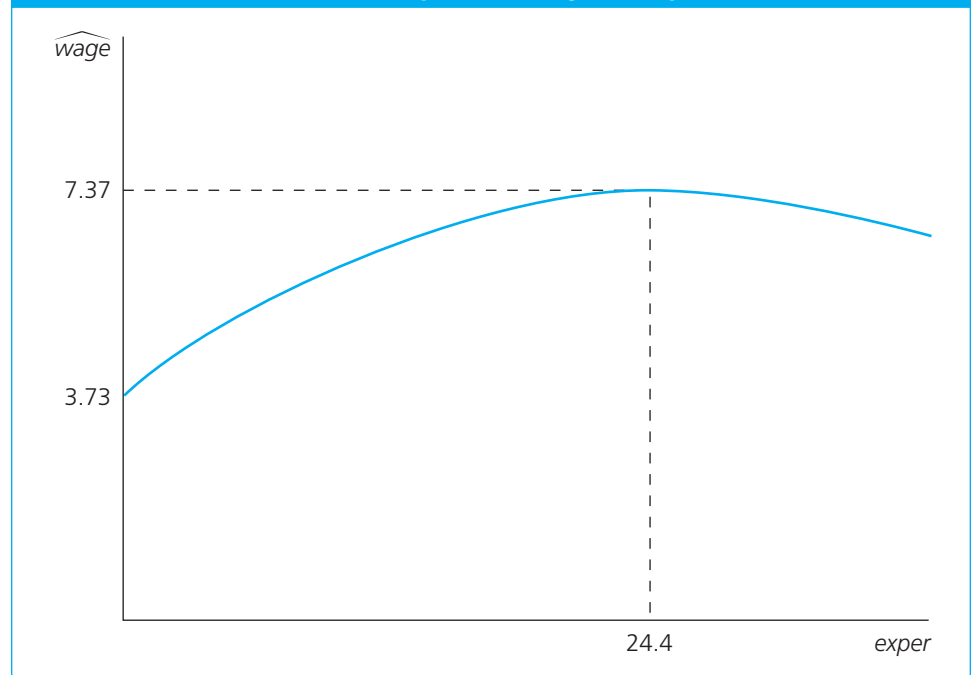
In the estimated equation (6.10) with $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$, the turning point (or maximum of the function) is always achieved at the coefficient on x over *twice* the absolute value of the coefficient on x^2 :

$$x^* = |\hat{\beta}_1 / (2\hat{\beta}_2)|. \quad [6.13]$$

In the wage example, $x^* = \text{exper}^*$ is $.298 / [2(.0061)] \approx 24.4$. (Note how we just drop the minus sign on $-.0061$ in doing this calculation.) This quadratic relationship is illustrated in Figure 6.1.

In the wage equation (6.12), the return to experience becomes zero at about 24.4 years. What should we make of this? There are at least three possible explanations. First, it may be that few people in the sample have more than 24 years of experience, and so the part of the curve to the right of 24 can be ignored. The cost of using a quadratic to capture diminishing effects is that the quadratic must eventually turn around. If this point is beyond all but a small percentage of the people in the sample, then this is not of much concern.

FIGURE 6.1 Quadratic relationship between $\widehat{\text{wage}}$ and *exper*.



But in the data set WAGE1.RAW, about 28% of the people in the sample have more than 24 years of experience; this is too high a percentage to ignore.

It is possible that the return to *exper* really becomes negative at some point, but it is hard to believe that this happens at 24 years of experience. A more likely possibility is that the estimated effect of *exper* on *wage* is biased because we have controlled for no other factors, or because the functional relationship between *wage* and *exper* in equation (6.12) is not entirely correct. Computer Exercise C2 asks you to explore this possibility by controlling for education, in addition to using $\log(\text{wage})$ as the dependent variable.

When a model has a dependent variable in logarithmic form and an explanatory variable entering as a quadratic, some care is needed in reporting the partial effects. The following example also shows that the quadratic can have a U-shape, rather than a parabolic shape. A U-shape arises in equation (6.10) when $\hat{\beta}_1$ is negative and $\hat{\beta}_2$ is positive; this captures an increasing effect of x on y .

EXAMPLE 6.2

EFFECTS OF POLLUTION ON HOUSING PRICES

We modify the housing price model from Example 4.5 to include a quadratic term in *rooms*:

$$\begin{aligned}\log(\text{price}) = & \beta_0 + \beta_1 \log(\text{nox}) + \beta_2 \log(\text{dist}) + \beta_3 \text{rooms} \\ & + \beta_4 \text{rooms}^2 + \beta_5 \text{stratio} + u.\end{aligned}\quad [6.14]$$

The model estimated using the data in HPRICE2.RAW is

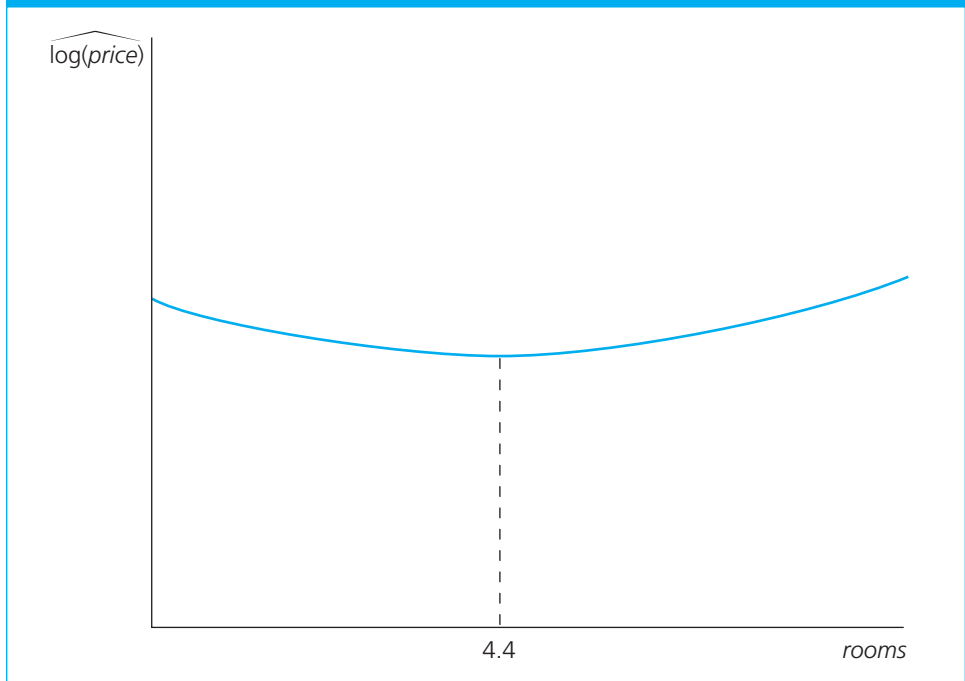
$$\begin{aligned}\widehat{\log(\text{price})} = & 13.39 - .902 \log(\text{nox}) - .087 \log(\text{dist}) \\ & (.57) \quad (.115) \quad (.043) \\ & - .545 \text{rooms} + .062 \text{rooms}^2 - .048 \text{stratio} \\ & (.165) \quad (.013) \quad (.006) \\ n = & 506, R^2 = .603.\end{aligned}$$

The quadratic term rooms^2 has a t statistic of about 4.77, and so it is very statistically significant. But what about interpreting the effect of *rooms* on $\log(\text{price})$? Initially, the effect appears to be strange. Because the coefficient on *rooms* is negative and the coefficient on rooms^2 is positive, this equation literally implies that, at low values of *rooms*, an additional room has a *negative* effect on $\log(\text{price})$. At some point, the effect becomes positive, and the quadratic shape means that the semi-elasticity of *price* with respect to *rooms* is increasing as *rooms* increases. This situation is shown in Figure 6.2.

We obtain the turnaround value of *rooms* using equation (6.13) (even though $\hat{\beta}_1$ is negative and $\hat{\beta}_2$ is positive). The absolute value of the coefficient on *rooms*, .545, divided by twice the coefficient on rooms^2 , .062, gives $\text{rooms}^* = .545/[2(.062)] \approx 4.4$; this point is labeled in Figure 6.2.

Do we really believe that starting at three rooms and increasing to four rooms actually reduces a house's expected value? Probably not. It turns out that only five of the 506 communities in the sample have houses averaging 4.4 rooms or less, about 1% of the sample. This is so small that the quadratic to the left of 4.4 can, for practical purposes, be ignored. To the right of 4.4, we see that adding another room has an increasing effect on the percentage change in price:

$$\Delta \widehat{\log(\text{price})} \approx \{[-.545 + 2(.062)]\text{rooms}\} \Delta \text{rooms}$$

FIGURE 6.2 $\widehat{\log(\text{price})}$ as a quadratic function of *rooms*.

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and so

$$\begin{aligned}\% \Delta \widehat{\text{price}} &\approx 100\{[-.545 + 2(.062)]\text{rooms}\} \Delta \text{rooms} \\ &= (-54.5 + 12.4 \text{ rooms}) \Delta \text{rooms}.\end{aligned}$$

Thus, an increase in *rooms* from, say, five to six increases price by about $-54.5 + 12.4(5) = 7.5\%$; the increase from six to seven increases price by roughly $-54.5 + 12.4(6) = 19.9\%$. This is a very strong increasing effect.

The strong increasing effect of *rooms* on $\log(\text{price})$ in this example illustrates an important lesson: one cannot simply look at the coefficient on the quadratic term—in this case, .062—and declare that it is too small to bother with based only on its magnitude. In many applications with quadratics the coefficient on the squared variable has one or more zeros after the decimal point: after all, this coefficient measures how the slope is changing as x (*rooms*) changes. A seemingly small coefficient can have practically important consequences, as we just saw. As a general rule, one must compute the partial effect and see how it varies with x to determine if the quadratic term is practically important. In doing so, it is useful to compare the changing slope implied by the quadratic model with the constant slope obtained from the model with only a linear term. If we drop rooms^2 from the equation, the coefficient on *rooms* becomes about .255, which implies that each additional room—starting from any number of rooms—increases median price by about 25.5%. This is very different from the quadratic model, where the effect becomes 25.5% at $\text{rooms} = 6.45$ but changes rapidly as *rooms* gets smaller or larger. For example, at $\text{rooms} = 7$, the return to the next room is about 32.3%.

What happens generally if the coefficients on the level and squared terms have the *same* sign (either both positive or both negative) and the explanatory variable is necessarily nonnegative (as in the case of *rooms* or *exper*)? In either case, there is no turning point for values $x > 0$. For example, if β_1 and β_2 are both positive, the smallest expected value of y is at $x = 0$, and increases in x always have a positive and increasing effect on y . (This is also true if $\beta_1 = 0$ and $\beta_2 > 0$, which means that the partial effect is zero at $x = 0$ and increasing as x increases.) Similarly, if β_1 and β_2 are both negative, the largest expected value of y is at $x = 0$, and increases in x have a negative effect on y , with the magnitude of the effect increasing as x gets larger.

The general formula for the turning point of any quadratic is $x^* = -\hat{\beta}_1/(2\hat{\beta}_2)$, which leads to a positive value if $\hat{\beta}_1$ and $\hat{\beta}_2$ have opposite signs and a negative value when $\hat{\beta}_1$ and $\hat{\beta}_2$ have the same sign. Knowing this simple formula is useful in cases where x may take on both positive and negative values; one can compute the turning point and see if it makes sense, taking into account the range of x in the sample.

There are many other possibilities for using quadratics along with logarithms. For example, an extension of (6.14) that allows a nonconstant elasticity between *price* and *nox* is

$$\log(\text{price}) = \beta_0 + \beta_1 \log(\text{nox}) + \beta_2 [\log(\text{nox})]^2 + \beta_3 \text{crime} + \beta_4 \text{rooms} + \beta_5 \text{rooms}^2 + \beta_6 \text{stratio} + u. \quad [6.15]$$

If $\beta_2 = 0$, then β_1 is the elasticity of *price* with respect to *nox*. Otherwise, this elasticity depends on the level of *nox*. To see this, we can combine the arguments for the partial effects in the quadratic and logarithmic models to show that

$$\% \Delta \text{price} \approx [\beta_1 + 2\beta_2 \log(\text{nox})] \% \Delta \text{nox}; \quad [6.16]$$

therefore, the elasticity of *price* with respect to *nox* is $\beta_1 + 2\beta_2 \log(\text{nox})$, so that it depends on $\log(\text{nox})$.

Finally, other polynomial terms can be included in regression models. Certainly, the quadratic is seen most often, but a cubic and even a quartic term appear now and then. An often reasonable functional form for a total cost function is

$$\text{cost} = \beta_0 + \beta_1 \text{quantity} + \beta_2 \text{quantity}^2 + \beta_3 \text{quantity}^3 + u.$$

Estimating such a model causes no complications. Interpreting the parameters is more involved (though straightforward using calculus); we do not study these models further.

Models with Interaction Terms

Sometimes, it is natural for the partial effect, elasticity, or semi-elasticity of the dependent variable with respect to an explanatory variable to depend on the magnitude of yet *another* explanatory variable. For example, in the model

$$\text{price} = \beta_0 + \beta_1 \text{sqrft} + \beta_2 \text{bdrms} + \beta_3 \text{sqrft} \cdot \text{bdrms} + \beta_4 \text{bthrms} + u,$$

the partial effect of *bdrms* on *price* (holding all other variables fixed) is

$$\frac{\Delta \text{price}}{\Delta \text{bdrms}} = \beta_2 + \beta_3 \text{sqrft}. \quad [6.17]$$

If $\beta_3 > 0$, then (6.17) implies that an additional bedroom yields a higher increase in housing price for larger houses. In other words, there is an **interaction effect** between square footage and number of bedrooms. In summarizing the effect of *bdrms* on *price*, we must evaluate (6.17) at interesting values of *sqrft*, such as the mean value, or the lower and upper quartiles in the sample. Whether or not β_3 is zero is something we can easily test.

The parameters on the original variables can be tricky to interpret when we include an interaction term. For example, in the previous housing price equation, equation (6.17) shows that β_2 is the effect of *bdrms* on *price* for a home with zero square feet! This effect is clearly not of much interest. Instead, we must be careful to put interesting values of *sqrft*, such as the mean or median values in the sample, into the estimated version of equation (6.17).

Often, it is useful to reparameterize a model so that the coefficients on the original variables have an interesting meaning. Consider a model with two explanatory variables and an interaction:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u.$$

As just mentioned, β_2 is the partial effect of x_2 on y when $x_1 = 0$. Often, this is not of interest. Instead, we can reparameterize the model as

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u,$$

where μ_1 is the population mean of x_1 and μ_2 is the population mean of x_2 . We can easily see that now the coefficient on x_2 , δ_2 , is the partial effect of x_2 on y at the mean value of x_1 . (By multiplying out the interaction in the second equation and comparing the coefficients, we can easily show that $\delta_2 = \beta_2 + \beta_3 \mu_1$. The parameter δ_1 has a similar interpretation.) Therefore, if we subtract the means of the variables—in practice, these would typically be the sample means—before creating the interaction term, the coefficients on the original variables have a useful interpretation. Plus, we immediately obtain standard errors for the partial effects at the mean values. Nothing prevents us from replacing μ_1 or μ_2 with other values of the explanatory variables that may be of interest. The following example illustrates how we can use interaction terms.

EXAMPLE 6.3

EFFECTS OF ATTENDANCE ON FINAL EXAM PERFORMANCE

A model to explain the standardized outcome on a final exam (*stndfnl*) in terms of percentage of classes attended, prior college grade point average, and ACT score is

$$\begin{aligned} \text{stndfnl} = & \beta_0 + \beta_1 \text{atndrte} + \beta_2 \text{priGPA} + \beta_3 \text{ACT} + \beta_4 \text{priGPA}^2 \\ & + \beta_5 \text{ACT}^2 + \beta_6 \text{priGPA} \cdot \text{atndrte} + u. \end{aligned}$$

[6.18]

(We use the standardized exam score for the reasons discussed in Section 6.1: it is easier to interpret a student's performance relative to the rest of the class.) In addition to quadratics in *priGPA* and *ACT*, this model includes an interaction between *priGPA* and the attendance rate. The idea is that class attendance might have a different effect for students who have performed differently in the past, as measured by *priGPA*. We are interested in the effects of attendance on final exam score: $\Delta \text{stndfnl} / \Delta \text{atndrte} = \beta_1 + \beta_6 \text{priGPA}$.

Using the 680 observations in ATTEND.RAW, for students in a course on microeconomic principles, the estimated equation is

$$\begin{aligned}\widehat{stndfnl} = & 2.05 - .0067 \textit{atndrte} - 1.63 \textit{priGPA} - .128 \textit{ACT} \\ & (1.36) \quad (.0102) \quad \quad (.48) \quad \quad (.098) \\ & + .296 \textit{priGPA}^2 + .0045 \textit{ACT}^2 + .0056 \textit{priGPA} \cdot \textit{atndrte} \quad [6.19] \\ & (.101) \quad \quad (.0022) \quad \quad (.0043) \\ n = & 680, R^2 = .229, \bar{R}^2 = .222.\end{aligned}$$

We must interpret this equation with extreme care. If we simply look at the coefficient on *atndrte*, we will incorrectly conclude that attendance has a *negative* effect on final exam score. But this coefficient supposedly measures the effect when *priGPA* = 0, which is not interesting (in this sample, the smallest prior GPA is about .86). We must also take care not to look separately at the estimates of β_1 and β_6 and conclude that, because each *t* statistic is insignificant, we cannot reject $H_0: \beta_1 = 0, \beta_6 = 0$. In fact, the *p*-value for the *F* test of this joint hypothesis is .014, so we certainly reject H_0 at the 5% level. This is a good example of where looking at separate *t* statistics when testing a joint hypothesis can lead one far astray.

How should we estimate the partial effect of *atndrte* on *stndfnl*? We must plug in interesting values of *priGPA* to obtain the partial effect. The mean value of *priGPA* in the sample is 2.59, so at the mean *priGPA*, the effect of *atndrte* on *stndfnl* is $-.0067 + .0056(2.59) \approx .0078$. What does this mean? Because *atndrte* is measured as a percentage, it means that a 10 percentage point increase in *atndrte* increases *stndfnl* by .078 standard deviations from the mean final exam score.

EXPLORING FURTHER 6.3

If we add the term $\beta_7 \textit{ACT} \cdot \textit{atndrte}$ to equation (6.18), what is the partial effect of *atndrte* on *stndfnl*?

How can we tell whether the estimate .0078 is statistically different from zero? We need to rerun the regression, where we replace *priGPA* · *atndrte* with $(\textit{priGPA} - 2.59) \cdot \textit{atndrte}$. This gives, as the new coefficient on *atndrte*, the estimated effect

at *priGPA* = 2.59, along with its standard error; nothing else in the regression changes. (We described this device in Section 4.4.) Running this new regression gives the standard error of $\hat{\beta}_1 + \hat{\beta}_6(2.59) = .0078$ as .0026, which yields $t = .0078/.0026 = 3$. Therefore, at the average *priGPA*, we conclude that attendance has a statistically significant positive effect on final exam score.

Things are even more complicated for finding the effect of *priGPA* on *stndfnl* because of the quadratic term *priGPA*². To find the effect at the mean value of *priGPA* and the mean attendance rate, 82, we would replace *priGPA*² with $(\textit{priGPA} - 2.59)^2$ and *priGPA* · *atndrte* with *priGPA* · (*atndrte* - 82). The coefficient on *priGPA* becomes the partial effect at the mean values, and we would have its standard error. (See Computer Exercise C7.)

6.3 More on Goodness-of-Fit and Selection of Regressors

Until now, we have not focused much on the size of R^2 in evaluating our regression models, because beginning students tend to put too much weight on *R*-squared. As we will see shortly, choosing a set of explanatory variables based on the size of the *R*-squared can lead to nonsensical models. In Chapter 10, we will discover that *R*-squareds obtained from time series regressions can be artificially high and can result in misleading conclusions.

Nothing about the classical linear model assumptions requires that R^2 be above any particular value; R^2 is simply an estimate of how much variation in y is explained by $x_1, x_2, \dots, x_k$ in the population. We have seen several regressions that have had pretty small R -squareds. Although this means that we have not accounted for several factors that affect y , this does not mean that the factors in u are correlated with the independent variables. The zero conditional mean assumption MLR.4 is what determines whether we get unbiased estimators of the ceteris paribus effects of the independent variables, and the size of the R -squared has no direct bearing on this.

A small R -squared does imply that the error variance is large relative to the variance of y , which means we may have a hard time precisely estimating the β_j . But remember, we saw in Section 3.4 that a large error variance can be offset by a large sample size: if we have enough data, we may be able to precisely estimate the partial effects even though we have not controlled for many unobserved factors. Whether or not we can get precise enough estimates depends on the application. For example, suppose that some incoming students at a large university are randomly given grants to buy computer equipment. If the amount of the grant is truly randomly determined, we can estimate the ceteris paribus effect of the grant amount on subsequent college grade point average by using simple regression analysis. (Because of random assignment, all of the other factors that affect GPA would be uncorrelated with the amount of the grant.) It seems likely that the grant amount would explain little of the variation in GPA, so the R -squared from such a regression would probably be very small. But, if we have a large sample size, we still might get a reasonably precise estimate of the effect of the grant.

Another good illustration of where poor explanatory power has nothing to do with unbiased estimation of the β_j is given by analyzing the data set APPLE.RAW. Unlike the other data sets we have used, the key explanatory variables in APPLE.RAW were set experimentally—that is, without regard to other factors that might affect the dependent variable. The variable we would like to explain, *ecolbs*, is the (hypothetical) pounds of “ecologically friendly” (“ecolabeled”) apples a family would demand. Each family (actually, family head) was presented with a description of ecolabeled apples, along with prices of regular apples (*regprc*) and prices of the hypothetical ecolabeled apples (*ecopr*). Because the price pairs were randomly assigned to each family, they are unrelated to other observed factors (such as family income) and unobserved factors (such as desire for a clean environment). Therefore, the regression of *ecolbs* on *ecopr*, *regprc* (across all samples generated in this way) produces unbiased estimators of the price effects. Nevertheless, the R -squared from the regression is only .0364: the price variables explain only about 3.6% of the total variation in *ecolbs*. So, here is a case where we explain very little of the variation in y , yet we are in the rare situation of knowing that the data have been generated so that unbiased estimation of the β_j is possible. (Incidentally, adding observed family characteristics has a very small effect on explanatory power. See Computer Exercise C11.)

Remember, though, that the relative *change* in the R -squared when variables are added to an equation is very useful: the F statistic in (4.41) for testing the joint significance crucially depends on the difference in R -squareds between the unrestricted and restricted models.

As we will see in Section 6.4, an important consequence of a low R -squared is that prediction is difficult. Because most of the variation in y is explained by unobserved factors (or at least factors we do not include in our model), we will generally have a hard time using the OLS equation to predict future outcomes on y given a set of values for the explanatory variables.

Adjusted R -Squared

Most regression packages will report, along with the R -squared, a statistic called the **adjusted R -squared**. Because the adjusted R -squared is reported in much applied work, and because it has some useful features, we cover it in this subsection.

To see how the usual R -squared might be adjusted, it is usefully written as

$$R^2 = 1 - (\text{SSR}/n)/(\text{SST}/n), \quad [6.20]$$

where SSR is the sum of squared residuals and SST is the total sum of squares; compared with equation (3.28), all we have done is divide both SSR and SST by n . This expression reveals what R^2 is actually estimating. Define σ_y^2 as the population variance of y and let σ_u^2 denote the population variance of the error term, u . (Until now, we have used σ^2 to denote σ_u^2 , but it is helpful to be more specific here.) The **population R -squared** is defined as $\rho^2 = 1 - \sigma_u^2/\sigma_y^2$; this is the proportion of the variation in y in the population explained by the independent variables. This is what R^2 is supposed to be estimating.

R^2 estimates σ_u^2 by SSR/n , which we know to be biased. So why not replace SSR/n with $\text{SSR}/(n - k - 1)$? Also, we can use $\text{SST}/(n - 1)$ in place of SST/n , as the former is the unbiased estimator of σ_y^2 . Using these estimators, we arrive at the adjusted R -squared:

$$\begin{aligned} \bar{R}^2 &= 1 - [\text{SSR}/(n - k - 1)]/[\text{SST}/(n - 1)] \\ &= 1 - \hat{\sigma}^2/[\text{SST}/(n - 1)], \end{aligned} \quad [6.21]$$

because $\hat{\sigma}^2 = \text{SSR}/(n - k - 1)$. Because of the notation used to denote the adjusted R -squared, it is sometimes called *R -bar squared*.

The adjusted R -squared is sometimes called the *corrected R -squared*, but this is not a good name because it implies that $\bar{R}^2$ is somehow better than R^2 as an estimator of the population R -squared. Unfortunately, $\bar{R}^2$ is *not* generally known to be a better estimator. It is tempting to think that $\bar{R}^2$ corrects the bias in R^2 for estimating the population R -squared, ρ^2 , but it does not: the ratio of two unbiased estimators is not an unbiased estimator.

The primary attractiveness of $\bar{R}^2$ is that it imposes a penalty for adding additional independent variables to a model. We know that R^2 can never fall when a new independent variable is added to a regression equation: this is because SSR never goes up (and usually falls) as more independent variables are added. But the formula for $\bar{R}^2$ shows that it depends explicitly on k , the number of independent variables. If an independent variable is added to a regression, SSR falls, but so does the df in the regression, $n - k - 1$. $\text{SSR}/(n - k - 1)$ can go up or down when a new independent variable is added to a regression.

An interesting algebraic fact is the following: If we add a new independent variable to a regression equation, $\bar{R}^2$ increases if, and only if, the t statistic on the new variable is greater than one in absolute value. (An extension of this is that $\bar{R}^2$ increases when a group of variables is added to a regression if, and only if, the F statistic for joint significance of the new variables is greater than unity.) Thus, we see immediately that using $\bar{R}^2$ to decide whether a certain independent variable (or set of variables) belongs in a model gives us a different answer than standard t or F testing (because a t or F statistic of unity is not statistically significant at traditional significance levels).

It is sometimes useful to have a formula for $\bar{R}^2$ in terms of R^2 . Simple algebra gives

$$\bar{R}^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1). \quad [6.22]$$

For example, if $R^2 = .30$, $n = 51$, and $k = 10$, then $\bar{R}^2 = 1 - .70(50)/40 = .125$. Thus, for small n and large k , $\bar{R}^2$ can be substantially below R^2 . In fact, if the usual R -squared is small, and $n - k - 1$ is small, $\bar{R}^2$ can actually be negative! For example, you can plug in $R^2 = .10$, $n = 51$, and $k = 10$ to verify that $\bar{R}^2 = -.125$. A negative $\bar{R}^2$ indicates a very poor model fit relative to the number of degrees of freedom.

The adjusted R -squared is sometimes reported along with the usual R -squared in regressions, and sometimes $\bar{R}^2$ is reported in place of R^2 . It is important to remember that it is R^2 , not $\bar{R}^2$, that appears in the F statistic in (4.41). The same formula with $\bar{R}_r^2$ and $\bar{R}_{ur}^2$ is *not* valid.

Using Adjusted R -Squared to Choose between Nonnested Models

In Section 4.5, we learned how to compute an F statistic for testing the joint significance of a group of variables; this allows us to decide, at a particular significance level, whether at least one variable in the group affects the dependent variable. This test does not allow us to decide *which* of the variables has an effect. In some cases, we want to choose a model without redundant independent variables, and the adjusted R -squared can help with this.

In the major league baseball salary example in Section 4.5, we saw that neither *hrunsyr* nor *rbisyr* was individually significant. These two variables are highly correlated, so we might want to choose between the models

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + u$$

and

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{rbisyr} + u.$$

These two equations are **nonnested models** because neither equation is a special case of the other. The F statistics we studied in Chapter 4 only allow us to test *nested* models: one model (the restricted model) is a special case of the other model (the unrestricted model). See equations (4.32) and (4.28) for examples of restricted and unrestricted models. One possibility is to create a composite model that contains *all* explanatory variables from the original models and then to test each model against the general model using the F test. The problem with this process is that either both models might be rejected or neither model might be rejected (as happens with the major league baseball salary example in Section 4.5). Thus, it does not always provide a way to distinguish between models with nonnested regressors.

In the baseball player salary regression, $\bar{R}^2$ for the regression containing *hrunsyr* is .6211, and $\bar{R}^2$ for the regression containing *rbisyr* is .6226. Thus, based on the adjusted R -squared, there is a very slight preference for the model with *rbisyr*. But the difference is practically very small, and we might obtain a different answer by controlling for some of the variables in Computer Exercise C5 in Chapter 4. (Because both nonnested models contain five parameters, the usual R -squared can be used to draw the same conclusion.)

Comparing $\bar{R}^2$ to choose among different nonnested sets of independent variables can be valuable when these variables represent different functional forms. Consider two models relating R&D intensity to firm sales:

$$rdintens = \beta_0 + \beta_1 \log(\text{sales}) + u. \quad [6.23]$$

$$rdintens = \beta_0 + \beta_1 \text{sales} + \beta_2 \text{sales}^2 + u. \quad [6.24]$$

The first model captures a diminishing return by including *sales* in logarithmic form; the second model does this by using a quadratic. Thus, the second model contains one more parameter than the first.

When equation (6.23) is estimated using the 32 observations on chemical firms in RDCHEM.RAW, R^2 is .061, and R^2 for equation (6.24) is .148. Therefore, it appears that the quadratic fits much better. But a comparison of the usual R -squareds is unfair to the first model because it contains one fewer parameter than (6.24). That is, (6.23) is a more parsimonious model than (6.24).

Everything else being equal, simpler models are better. Since the usual R -squared does not penalize more complicated models, it is better to use $\bar{R}^2$. $\bar{R}^2$ for (6.23) is .030, while $\bar{R}^2$ for (6.24) is .090. Thus, even after adjusting for the difference in degrees of freedom, the quadratic model wins out. The quadratic model is also preferred when profit margin is added to each regression.

There is an important limitation in using $\bar{R}^2$ to choose between nonnested models: we cannot use it to choose between different functional forms for the dependent variable. This is unfortunate, because we often want to decide on whether y or $\log(y)$ (or maybe some other transformation) should be used as the dependent variable based on goodness-of-fit. But neither R^2 nor $\bar{R}^2$ can be used for this purpose. The reason is simple: these R -squareds measure the explained proportion of the total variation in whatever dependent variable we are using in the regression, and different functions of the dependent variable will have different amounts of variation to explain. For example, the total variations in y and $\log(y)$ are not the same, and are often very different. Comparing the adjusted R -squareds from

regressions with these different forms of the dependent variables does not tell us anything about which model fits better; they are fitting two separate dependent variables.

EXPLORING FURTHER 6.4

Explain why choosing a model by maximizing $\bar{R}^2$ or minimizing $\hat{\sigma}$ (the standard error of the regression) is the same thing.

EXAMPLE 6.4

CEO COMPENSATION AND FIRM PERFORMANCE

Consider two estimated models relating CEO compensation to firm performance:

$$\begin{aligned} \widehat{\text{salary}} &= 830.63 + .0163 \text{ sales} + 19.63 \text{ roe} \\ &\quad (223.90) \quad (.0089) \quad (11.08) \end{aligned} \quad [6.25]$$

$$n = 209, R^2 = .029, \bar{R}^2 = .020$$

and

$$\begin{aligned} \widehat{\ln \text{salary}} &= 4.36 + .275 \ln \text{sales} + .0179 \text{ roe} \\ &\quad (0.29) \quad (.033) \quad (.0040) \end{aligned} \quad [6.26]$$

$$n = 209, R^2 = .282, \bar{R}^2 = .275,$$

where *roe* is the return on equity discussed in Chapter 2. For simplicity, *lsalary* and *lsales* denote the natural logs of *salary* and *sales*. We already know how to interpret these different estimated equations. But can we say that one model fits better than the other?

The R -squared for equation (6.25) shows that *sales* and *roe* explain only about 2.9% of the variation in CEO salary in the sample. Both *sales* and *roe* have marginal statistical significance.

Equation (6.26) shows that $\log(\text{sales})$ and roe explain about 28.2% of the variation in $\log(\text{salary})$. In terms of goodness-of-fit, this much higher R -squared would seem to imply that model (6.26) is much better, but this is not necessarily the case. The total sum of squares for salary in the sample is 391,732,982, while the total sum of squares for $\log(\text{salary})$ is only 66.72. Thus, there is much less variation in $\log(\text{salary})$ that needs to be explained.

At this point, we can use features other than R^2 or $\bar{R}^2$ to decide between these models. For example, $\log(\text{sales})$ and roe are much more statistically significant in (6.26) than are sales and roe in (6.25), and the coefficients in (6.26) are probably of more interest. To be sure, however, we will need to make a valid goodness-of-fit comparison.

In Section 6.4, we will offer a goodness-of-fit measure that does allow us to compare models where y appears in both level and log form.

Controlling for Too Many Factors in Regression Analysis

In many of the examples we have covered, and certainly in our discussion of omitted variables bias in Chapter 3, we have worried about omitting important factors from a model that might be correlated with the independent variables. It is also possible to control for too *many* variables in a regression analysis.

If we overemphasize goodness-of-fit, we open ourselves to controlling for factors in a regression model that should not be controlled for. To avoid this mistake, we need to remember the *ceteris paribus* interpretation of multiple regression models.

To illustrate this issue, suppose we are doing a study to assess the impact of state beer taxes on traffic fatalities. The idea is that a higher tax on beer will reduce alcohol consumption, and likewise drunk driving, resulting in fewer traffic fatalities. To measure the *ceteris paribus* effect of taxes on fatalities, we can model *fatalities* as a function of several factors, including the beer *tax*:

$$\text{fatalities} = \beta_0 + \beta_1 \text{tax} + \beta_2 \text{miles} + \beta_3 \text{percmales} + \beta_4 \text{perc16_21} + \dots,$$

where

miles = total miles driven.

percmales = percentage of the state population that is male.

perc16_21 = percentage of the population between ages 16 and 21, and so on.

Notice how we have not included a variable measuring per capita beer consumption. Are we committing an omitted variables error? The answer is no. If we control for beer consumption in this equation, then how would beer taxes affect traffic fatalities? In the equation

$$\text{fatalities} = \beta_0 + \beta_1 \text{tax} + \beta_2 \text{beercons} + \dots,$$

β_1 measures the difference in fatalities due to a one percentage point increase in *tax*, holding *beercons* fixed. It is difficult to understand why this would be interesting. We should not be controlling for differences in *beercons* across states, unless we want to test for some sort of indirect effect of beer taxes. Other factors, such as gender and age distribution, should be controlled for.

As a second example, suppose that, for a developing country, we want to estimate the effects of pesticide usage among farmers on family health expenditures. In addition to pesticide usage amounts, should we include the number of doctor visits as an explanatory variable? No. Health expenditures include doctor visits, and we would like to pick up all effects of pesticide use on health expenditures. If we include the number of doctor visits as an explanatory variable, then we are only measuring the effects of pesticide use on health expenditures other than doctor visits. It makes more sense to use number of doctor visits as a dependent variable in a separate regression on pesticide amounts.

The previous examples are what can be called **over controlling** for factors in multiple regression. Often this results from nervousness about potential biases that might arise by leaving out an important explanatory variable. But it is important to remember the *ceteris paribus* nature of multiple regression. In some cases, it makes no sense to hold some factors fixed precisely because they should be allowed to change when a policy variable changes.

Unfortunately, the issue of whether or not to control for certain factors is not always clear-cut. For example, Betts (1995) studies the effect of high school quality on subsequent earnings. He points out that, if better school quality results in more education, then controlling for education in the regression along with measures of quality will underestimate the return to quality. Betts does the analysis with and without years of education in the equation to get a range of estimated effects for quality of schooling.

To see explicitly how pursuing high *R*-squareds can lead to trouble, consider the housing price example from Section 4.5 that illustrates the testing of multiple hypotheses. In that case, we wanted to test the rationality of housing price assessments. We regressed $\log(\text{price})$ on $\log(\text{assess})$, $\log(\text{lotsize})$, $\log(\text{sqrft})$, and bdrms and tested whether the latter three variables had zero population coefficients while $\log(\text{assess})$ had a coefficient of unity. But what if we change the purpose of the analysis and estimate a *hedonic price model*, which allows us to obtain the marginal values of various housing attributes? Should we include $\log(\text{assess})$ in the equation? The adjusted *R*-squared from the regression with $\log(\text{assess})$ is .762, while the adjusted *R*-squared without it is .630. Based on goodness-of-fit only, we should include $\log(\text{assess})$. But this is incorrect if our goal is to determine the effects of lot size, square footage, and number of bedrooms on housing values. Including $\log(\text{assess})$ in the equation amounts to holding one measure of value fixed and then asking how much an additional bedroom would change another measure of value. This makes no sense for valuing housing attributes.

If we remember that different models serve different purposes, and we focus on the *ceteris paribus* interpretation of regression, then we will not include the wrong factors in a regression model.

Adding Regressors to Reduce the Error Variance

We have just seen some examples of where certain independent variables should not be included in a regression model, even though they are correlated with the dependent variable. From Chapter 3, we know that adding a new independent variable to a regression can exacerbate the multicollinearity problem. On the other hand, since we are taking something out of the error term, adding a variable generally reduces the error variance. Generally, we cannot know which effect will dominate.

However, there is one case that is clear: we should always include independent variables that affect *y* and are *uncorrelated* with all of the independent variables of interest. Why?

Because adding such a variable does not induce multicollinearity in the population (and therefore multicollinearity in the sample should be negligible), but it will reduce the error variance. In large sample sizes, the standard errors of all OLS estimators will be reduced.

As an example, consider estimating the individual demand for beer as a function of the average county beer price. It may be reasonable to assume that individual characteristics are uncorrelated with county-level prices, and so a simple regression of beer consumption on county price would suffice for estimating the effect of price on individual demand. But it is possible to get a more precise estimate of the price elasticity of beer demand by including individual characteristics, such as age and amount of education. If these factors affect demand and are uncorrelated with price, then the standard error of the price coefficient will be smaller, at least in large samples.

As a second example, consider the grants for computer equipment given at the beginning of Section 6.3. If, in addition to the grant variable, we control for other factors that can explain college GPA, we can probably get a more precise estimate of the effect of the grant. Measures of high school grade point average and rank, SAT and ACT scores, and family background variables are good candidates. Because the grant amounts are randomly assigned, all additional control variables are uncorrelated with the grant amount; in the sample, multicollinearity between the grant amount and other independent variables should be minimal. But adding the extra controls might significantly reduce the error variance, leading to a more precise estimate of the grant effect. Remember, the issue is not unbiasedness here: we obtain an unbiased and consistent estimator whether or not we add the high school performance and family background variables. The issue is getting an estimator with a smaller sampling variance.

Unfortunately, cases where we have information on additional explanatory variables that are uncorrelated with the explanatory variables of interest are somewhat rare in the social sciences. But it is worth remembering that when these variables are available, they can be included in a model to reduce the error variance without inducing multicollinearity.

6.4 Prediction and Residual Analysis

In Chapter 3, we defined the OLS predicted or fitted values and the OLS residuals. **Predictions** are certainly useful, but they are subject to sampling variation, because they are obtained using the OLS estimators. Thus, in this section, we show how to obtain confidence intervals for a prediction from the OLS regression line.

From Chapters 3 and 4, we know that the residuals are used to obtain the sum of squared residuals and the *R*-squared, so they are important for goodness-of-fit and testing. Sometimes, economists study the residuals for particular observations to learn about individuals (or firms, houses, etc.) in the sample.

Confidence Intervals for Predictions

Suppose we have estimated the equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k. \quad [6.27]$$

When we plug in particular values of the independent variables, we obtain a prediction for *y*, which is an estimate of the *expected value* of *y* given the particular values for the explanatory variables. For emphasis, let $c_1, c_2, \dots, c_k$ denote particular values for each of

the k independent variables; these may or may not correspond to an actual data point in our sample. The parameter we would like to estimate is

$$\begin{aligned}\theta_0 &= \beta_0 + \beta_1 c_1 + \beta_2 c_2 + \dots + \beta_k c_k \\ &= E(y | x_1 = c_1, x_2 = c_2, \dots, x_k = c_k).\end{aligned}\quad [6.28]$$

The estimator of θ_0 is

$$\hat{\theta}_0 = \hat{\beta}_0 + \hat{\beta}_1 c_1 + \hat{\beta}_2 c_2 + \dots + \hat{\beta}_k c_k. \quad [6.29]$$

In practice, this is easy to compute. But what if we want some measure of the uncertainty in this predicted value? It is natural to construct a confidence interval for θ_0 , which is centered at $\hat{\theta}_0$.

To obtain a confidence interval for θ_0 , we need a standard error for $\hat{\theta}_0$. Then, with a large df , we can construct a 95% confidence interval using the rule of thumb $\hat{\theta}_0 \pm 2 \cdot \text{se}(\hat{\theta}_0)$. (As always, we can use the exact percentiles in a t distribution.)

How do we obtain the standard error of $\hat{\theta}_0$? This is the same problem we encountered in Section 4.4: we need to obtain a standard error for a linear combination of the OLS estimators. Here, the problem is even more complicated, because all of the OLS estimators generally appear in $\hat{\theta}_0$ (unless some c_j are zero). Nevertheless, the same trick that we used in Section 4.4 will work here. Write $\beta_0 = \theta_0 - \beta_1 c_1 - \dots - \beta_k c_k$ and plug this into the equation

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$$

to obtain

$$y = \theta_0 + \beta_1(x_1 - c_1) + \beta_2(x_2 - c_2) + \dots + \beta_k(x_k - c_k) + u. \quad [6.30]$$

In other words, we subtract the value c_j from each observation on x_j , and then we run the regression of

$$y_i \text{ on } (x_{i1} - c_1), \dots, (x_{ik} - c_k), i = 1, 2, \dots, n. \quad [6.31]$$

The predicted value in (6.29) and, more importantly, its standard error, are obtained from the *intercept* (or constant) in regression (6.31).

As an example, we obtain a confidence interval for a prediction from a college GPA regression, where we use high school information.

EXAMPLE 6.5

CONFIDENCE INTERVAL FOR PREDICTED COLLEGE GPA

Using the data in GPA2.RAW, we obtain the following equation for predicting college GPA:

$$\begin{aligned}\widehat{colgpa} &= 1.493 + .00149 \text{ sat} - .01386 \text{ hspc} \\ &\quad (0.075) \quad (.00007) \quad (.00056) \\ &\quad - .06088 \text{ hsize} + .00546 \text{ hsize}^2 \\ &\quad (.01650) \quad (.00227) \\ n &= 4,137, R^2 = .278, \bar{R}^2 = .277, \hat{\sigma} = .560,\end{aligned}\quad [6.32]$$

where we have reported estimates to several digits to reduce round-off error. What is predicted college GPA, when $sat = 1,200$, $hsperc = 30$, and $hsize = 5$ (which means 500)? This is easy to get by plugging these values into equation (6.32): $\widehat{colgpa} = 2.70$ (rounded to two digits). Unfortunately, we cannot use equation (6.32) directly to get a confidence interval for the expected $colgpa$ at the given values of the independent variables. One simple way to obtain a confidence interval is to define a new set of independent variables: $sat0 = sat - 1,200$, $hsperc0 = hsperc - 30$, $hsize0 = hsize - 5$, and $hsizesq0 = hsize^2 - 25$. When we regress $colgpa$ on these new independent variables, we get

$$\begin{aligned}\widehat{colgpa} &= 2.700 + .00149 sat0 - .01386 hsperc0 \\ &\quad (0.020) \quad (.00007) \quad (.00056) \\ &\quad - .06088 hsize0 + .00546 hsizesq0 \\ &\quad (.01650) \quad (.00227) \\ n &= 4,137, R^2 = .278, \bar{R}^2 = .277, \hat{\sigma} = .560.\end{aligned}$$

The only difference between this regression and that in (6.32) is the intercept, which is the prediction we want, along with its standard error, .020. It is not an accident that the slope coefficients, their standard errors, R -squared, and so on are the same as before; this provides a way to check that the proper transformations were done. We can easily construct a 95% confidence interval for the expected college GPA: $2.70 \pm 1.96(.020)$ or about 2.66 to 2.74. This confidence interval is rather narrow due to the very large sample size.

Because the variance of the intercept estimator is smallest when each explanatory variable has zero sample mean (see Question 2.5 for the simple regression case), it follows from the regression in (6.31) that the variance of the prediction is smallest at the mean values of the x_j . (That is, $c_j = \bar{x}_j$ for all j .) This result is not too surprising, since we have the most faith in our regression line near the middle of the data. As the values of the c_j get farther away from the $\bar{x}_j$, $\text{Var}(\hat{y})$ gets larger and larger.

The previous method allows us to put a confidence interval around the OLS estimate of $E(y|x_1, \dots, x_k)$ for any values of the explanatory variables. In other words, we obtain a confidence interval for the *average* value of y for the subpopulation with a given set of covariates. But a confidence interval for the average person in the subpopulation is not the same as a confidence interval for a particular unit (individual, family, firm, and so on) from the population. In forming a confidence interval for an unknown outcome on y , we must account for another very important source of variation: the variance in the unobserved error, which measures our ignorance of the unobserved factors that affect y .

Let y^0 denote the value for which we would like to construct a confidence interval, which we sometimes call a **prediction interval**. For example, y^0 could represent a person or firm not in our original sample. Let $x_1^0, \dots, x_k^0$ be the new values of the independent variables, which we assume we observe, and let u^0 be the unobserved error. Therefore, we have

$$y^0 = \beta_0 + \beta_1 x_1^0 + \beta_2 x_2^0 + \dots + \beta_k x_k^0 + u^0. \quad [6.33]$$

As before, our best prediction of y^0 is the expected value of y^0 given the explanatory variables, which we estimate from the OLS regression line: $\hat{y}^0 = \hat{\beta}_0 + \hat{\beta}_1 x_1^0 + \hat{\beta}_2 x_2^0 + \dots + \hat{\beta}_k x_k^0$. The **prediction error** in using $\hat{y}^0$ to predict y^0 is

$$\hat{e}^0 = y^0 - \hat{y}^0 = (\beta_0 + \beta_1 x_1^0 + \dots + \beta_k x_k^0) + u^0 - \hat{y}^0. \quad [6.34]$$

Now, $E(\hat{y}^0) = E(\hat{\beta}_0) + E(\hat{\beta}_1)x_1^0 + E(\hat{\beta}_2)x_2^0 + \dots + E(\hat{\beta}_k)x_k^0 = \beta_0 + \beta_1x_1^0 + \dots + \beta_kx_k^0$, because the $\hat{\beta}_j$ are unbiased. (As before, these expectations are all conditional on the sample values of the independent variables.) Because u^0 has zero mean, $E(\hat{e}^0) = 0$. We have shown that the expected prediction error is zero.

In finding the variance of $\hat{e}^0$, note that u^0 is uncorrelated with each $\hat{\beta}_j$, because u^0 is uncorrelated with the errors in the sample used to obtain the $\hat{\beta}_j$. By basic properties of covariance (see Appendix B), u^0 and $\hat{y}^0$ are uncorrelated. Therefore, the **variance of the prediction error** (conditional on all in-sample values of the independent variables) is the sum of the variances:

$$\text{Var}(\hat{e}^0) = \text{Var}(\hat{y}^0) + \text{Var}(u^0) = \text{Var}(\hat{y}^0) + \sigma^2, \quad [6.35]$$

where $\sigma^2 = \text{Var}(u^0)$ is the error variance. There are two sources of variation in $\hat{e}^0$. The first is the sampling error in $\hat{y}^0$, which arises because we have estimated the β_j . Because each $\hat{\beta}_j$ has a variance proportional to $1/n$, where n is the sample size, $\text{Var}(\hat{y}^0)$ is proportional to $1/n$. This means that, for large samples, $\text{Var}(\hat{y}^0)$ can be very small. By contrast, σ^2 is the variance of the error in the population; it does not change with the sample size. In many examples, σ^2 will be the dominant term in (6.35).

Under the classical linear model assumptions, the $\hat{\beta}_j$ and u^0 are normally distributed, and so $\hat{e}^0$ is also normally distributed (conditional on all sample values of the explanatory variables). Earlier, we described how to obtain an unbiased estimator of $\text{Var}(\hat{y}^0)$, and we obtained our unbiased estimator of σ^2 in Chapter 3. By using these estimators, we can define the standard error of $\hat{e}^0$ as

$$\text{se}(\hat{e}^0) = \{[\text{se}(\hat{y}^0)]^2 + \hat{\sigma}^2\}^{1/2}. \quad [6.36]$$

Using the same reasoning for the t statistics of the $\hat{\beta}_j$, $\hat{e}^0/\text{se}(\hat{e}^0)$ has a t distribution with $n - (k + 1)$ degrees of freedom. Therefore,

$$P[-t_{.025} \leq \hat{e}^0/\text{se}(\hat{e}^0) \leq t_{.025}] = .95,$$

where $t_{.025}$ is the 97.5th percentile in the t_{n-k-1} distribution. For large $n - k - 1$, remember that $t_{.025} \approx 1.96$. Plugging in $\hat{e}^0 = y^0 - \hat{y}^0$ and rearranging gives a 95% prediction interval for y^0 :

$$\hat{y}^0 \pm t_{.025} \cdot \text{se}(\hat{e}^0); \quad [6.37]$$

as usual, except for small df , a good rule of thumb is $\hat{y}^0 \pm 2\text{se}(\hat{e}^0)$. This is wider than the confidence interval for $\hat{y}^0$ itself because of $\hat{\sigma}^2$ in (6.36); it often is much wider to reflect the factors in u^0 that we have not accounted for.

EXAMPLE 6.6

CONFIDENCE INTERVAL FOR FUTURE COLLEGE GPA

Suppose we want a 95% CI for the future college GPA of a high school student with $\text{sat} = 1,200$, $\text{hsperc} = 30$, and $\text{hsize} = 5$. In Example 6.5, we obtained a 95% confidence interval for the *average* college GPA among all students with the particular characteristics $\text{sat} = 1,200$, $\text{hsperc} = 30$, and $\text{hsize} = 5$. Now, we want a 95% confidence interval for any *particular* student with these characteristics. The 95% prediction interval must account for the variation in the individual, unobserved characteristics that affect college performance. We have everything we need to obtain a CI for colgpa . $\text{se}(\hat{y}^0) = .020$ and

$\hat{\sigma} = .560$ and so, from (6.36), $se(\hat{\epsilon}^0) = [(.020)^2 + (.560)^2]^{1/2} \approx .560$. Notice how small $se(\hat{y}^0)$ is relative to $\hat{\sigma}$: virtually all of the variation in $\hat{\epsilon}^0$ comes from the variation in u^0 . The 95% CI is $2.70 \pm 1.96(.560)$ or about 1.60 to 3.80. This is a wide confidence interval and shows that, based on the factors we included in the regression, we cannot accurately pin down an individual's future college grade point average. (In one sense, this is good news, as it means that high school rank and performance on the SAT do not preordain one's performance in college.) Evidently, the unobserved characteristics that affect college GPA vary widely among individuals with the same observed SAT score and high school rank.

Residual Analysis

Sometimes, it is useful to examine individual observations to see whether the actual value of the dependent variable is above or below the predicted value; that is, to examine the residuals for the individual observations. This process is called **residual analysis**. Economists have been known to examine the residuals from a regression in order to aid in the purchase of a home. The following housing price example illustrates residual analysis. Housing price is related to various observable characteristics of the house. We can list all of the characteristics that we find important, such as size, number of bedrooms, number of bathrooms, and so on. We can use a sample of houses to estimate a relationship between price and attributes, where we end up with a predicted value and an actual value for each house. Then, we can construct the residuals, $\hat{u}_i = y_i - \hat{y}_i$. The house with the most negative residual is, at least based on the factors we have controlled for, the most underpriced one relative to its *observed* characteristics. Of course, a selling price substantially below its predicted price could indicate some undesirable feature of the house that we have failed to account for, and which is therefore contained in the unobserved error. In addition to obtaining the prediction and residual, it also makes sense to compute a confidence interval for what the future selling price of the home could be, using the method described in equation (6.37).

Using the data in HPRICE1.RAW, we run a regression of *price* on *lotsize*, *sqrft*, and *bdrms*. In the sample of 88 homes, the most negative residual is -120.206 , for the 81st house. Therefore, the asking price for this house is \$120,206 below its predicted price.

There are many other uses of residual analysis. One way to rank law schools is to regress median starting salary on a variety of student characteristics (such as median LSAT scores of entering class, median college GPA of entering class, and so on) and to obtain a predicted value and residual for each law school. The law school with the largest residual has the highest predicted value added. (Of course, there is still much uncertainty about how an individual's starting salary would compare with the median for a law school overall.) These residuals can be used along with the costs of attending each law school to determine the best value; this would require an appropriate discounting of future earnings.

Residual analysis also plays a role in legal decisions. A *New York Times* article entitled "Judge Says Pupil's Poverty, Not Segregation, Hurts Scores" (6/28/95) describes an important legal case. The issue was whether the poor performance on standardized tests in the Hartford School District, relative to performance in surrounding suburbs, was due to poor school quality at the highly segregated schools. The judge concluded that "the disparity in test scores does not indicate that Hartford is doing an inadequate or poor job in educating its students or that its schools are failing, because the predicted scores based upon the relevant socioeconomic factors are about at the levels that one would expect." This conclusion is based on a regression analysis of average or median scores on

EXPLORING FURTHER 6.5

How would you use residual analysis to determine which professional athletes are overpaid or underpaid relative to their performance?

socioeconomic characteristics of various school districts in Connecticut. The judge's conclusion suggests that, given the poverty levels of students at Hartford schools, the actual test scores were similar to those predicted from a regression analysis: the residual for Hartford was not sufficiently negative to conclude that the schools themselves were the cause of low test scores.

Predicting y When $\log(y)$ Is the Dependent Variable

Because the natural log transformation is used so often for the dependent variable in empirical economics, we devote this subsection to the issue of predicting y when $\log(y)$ is the dependent variable. As a byproduct, we will obtain a goodness-of-fit measure for the log model that can be compared with the R -squared from the level model.

To obtain a prediction, it is useful to define $\log y = \log(y)$; this emphasizes that it is the log of y that is predicted in the model

$$\log y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u. \quad [6.38]$$

In this equation, the x_j might be transformations of other variables; for example, we could have $x_1 = \log(\text{sales})$, $x_2 = \log(\text{mktval})$, $x_3 = \text{ceoten}$ in the CEO salary example.

Given the OLS estimators, we know how to predict $\log y$ for any value of the independent variables:

$$\widehat{\log y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k. \quad [6.39]$$

Now, since the exponential undoes the log, our first guess for predicting y is to simply exponentiate the predicted value for $\log(y)$: $\hat{y} = \exp(\widehat{\log y})$. This does not work; in fact, it will systematically *underestimate* the expected value of y . In fact, if model (6.38) follows the CLM assumptions MLR.1 through MLR.6, it can be shown that

$$E(y|\mathbf{x}) = \exp(\sigma^2/2) \cdot \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k),$$

where $\mathbf{x}$ denotes the independent variables and σ^2 is the variance of u . [If $u \sim \text{Normal}(0, \sigma^2)$, then the expected value of $\exp(u)$ is $\exp(\sigma^2/2)$.] This equation shows that a simple adjustment is needed to predict y :

$$\hat{y} = \exp(\hat{\sigma}^2/2) \exp(\widehat{\log y}), \quad [6.40]$$

where $\hat{\sigma}^2$ is simply the unbiased estimator of σ^2 . Because $\hat{\sigma}$, the standard error of the regression, is always reported, obtaining predicted values for y is easy. Because $\hat{\sigma}^2 > 0$, $\exp(\hat{\sigma}^2/2) > 1$. For large $\hat{\sigma}^2$, this adjustment factor can be substantially larger than unity.

The prediction in (6.40) is not unbiased, but it is consistent. There are no unbiased predictions of y , and in many cases, (6.40) works well. However, it does rely on the normality of the error term, u . In Chapter 5, we showed that OLS has desirable properties, even when u is not normally distributed. Therefore, it is useful to have a prediction that does not rely on normality. If we just assume that u is independent of the explanatory variables, then we have

$$E(y|\mathbf{x}) = \alpha_0 \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k), \quad [6.41]$$

where α_0 is the expected value of $\exp(u)$, which must be greater than unity.

Given an estimate $\hat{\alpha}_0$, we can predict y as

$$\hat{y} = \hat{\alpha}_0 \exp(\widehat{\log y}), \quad [6.42]$$

which again simply requires exponentiating the predicted value from the log model and multiplying the result by $\hat{\alpha}_0$.

Two approaches suggest themselves for estimating α_0 without the normality assumption. The first is based on $\alpha_0 = E[\exp(u)]$. To estimate α_0 we replace the population expectation with a sample average and then we replace the unobserved errors, u_i , with the OLS residuals, $\hat{u}_i = \log(y_i) - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}$. This leads to the method of moments estimator (see Appendix C)

$$\hat{\alpha}_0 = n^{-1} \sum_{i=1}^n \exp(\hat{u}_i). \quad [6.43]$$

Not surprisingly, $\hat{\alpha}_0$ is a consistent estimator of α_0 , but it is not unbiased because we have replaced u_i with $\hat{u}_i$ inside a nonlinear function. This version of $\hat{\alpha}_0$ is a special case of what Duan (1983) called a **smearing estimate**. Because the OLS residuals have a zero sample average, it can be shown that, for any data set, $\hat{\alpha}_0 > 1$. (Technically, $\hat{\alpha}_0$ would equal one if all the OLS residuals were zero, but this never happens in any interesting application.) That $\hat{\alpha}_0$ is necessarily greater than one is convenient because it must be that $\alpha_0 > 1$.

A different estimate of α_0 is based on a simple regression through the origin. To see how it works, define $m_i = \exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik})$, so that, from equation (6.41), $E(y_i | m_i) = \alpha_0 m_i$. If we could observe the m_i , we could obtain an unbiased estimator of α_0 from the regression y_i on m_i without an intercept. Instead, we replace the β_j with their OLS estimates and obtain $\hat{m}_i = \exp(\widehat{\log y_i})$, where, of course, the $\widehat{\log y_i}$ are the fitted values from the regression $\log y_i$ on $x_{i1}, \dots, x_{ik}$ (with an intercept). Then $\check{\alpha}_0$ [to distinguish it from $\hat{\alpha}_0$ in equation (6.43)] is the OLS slope estimate from the simple regression y_i on $\hat{m}_i$ (no intercept):

$$\check{\alpha}_0 = \left(\sum_{i=1}^n \hat{m}_i^2 \right)^{-1} \left(\sum_{i=1}^n \hat{m}_i y_i \right). \quad [6.44]$$

We will call $\check{\alpha}_0$ the regression estimate of α_0 . Like $\hat{\alpha}_0$, $\check{\alpha}_0$ is consistent but not unbiased. Interestingly, $\check{\alpha}_0$ is not guaranteed to be greater than one, although it will be in most applications. If $\check{\alpha}_0$ is less than one, and especially if it is much less than one, it is likely that the assumption of independence between u and the x_j is violated. If $\check{\alpha}_0 < 1$, one possibility is to just use the estimate in (6.43), although this may simply be masking a problem with the linear model for $\log(y)$. We summarize the steps:

Predicting y When the Dependent Variable Is $\log(y)$:

1. Obtain the fitted values, $\widehat{\log y}_i$, and residuals, $\hat{u}_i$, from the regression $\log y$ on $x_1, \dots, x_k$.
2. Obtain $\hat{\alpha}_0$ as in equation (6.43) or $\check{\alpha}_0$ in equation (6.44).
3. For given values of $x_1, \dots, x_k$, obtain $\widehat{\log y}$ from (6.42).
4. Obtain the prediction $\hat{y}$ from (6.42) (with $\hat{\alpha}_0$ or $\check{\alpha}_0$).

We now show how to predict CEO salaries using this procedure.

EXAMPLE 6.7 **PREDICTING CEO SALARIES**

The model of interest is

$$\log(\text{salary}) = \beta_0 + \beta_1 \log(\text{sales}) + \beta_2 \log(\text{mktval}) + \beta_3 \text{ceoten} + u,$$

so that β_1 and β_2 are elasticities and $100 \cdot \beta_3$ is a semi-elasticity. The estimated equation using CEOSAL2.RAW is

$$\begin{aligned} \widehat{\log(\text{salary})} &= 4.504 + .163 \text{lsales} + .109 \text{lmktval} + .0117 \text{ceoten} \\ &\quad (.257) \quad (.039) \quad (.050) \quad (.0053) \quad [6.45] \\ n &= 177, R^2 = .318, \end{aligned}$$

where, for clarity, we let lsalary denote the log of salary , and similarly for lsales and lmktval . Next, we obtain $\hat{m}_i = \exp(\widehat{\log(\text{salary})}_i)$ for each observation in the sample.

The Duan smearing estimate from (6.43) is about $\hat{\alpha}_0 = 1.136$, and the regression estimate from (6.44) is $\check{\alpha}_0 = 1.117$. We can use either estimate to predict salary for any values of sales , mktval , and ceoten . Let us find the prediction for $\text{sales} = 5,000$ (which means \$5 billion because sales is in millions), $\text{mktval} = 10,000$ (or \$10 billion), and $\text{ceoten} = 10$. From (6.45), the prediction for lsalary is $4.504 + .163 \cdot \log(5,000) + .109 \cdot \log(10,000) + .0117(10) \approx 7.013$, and $\exp(7.013) \approx 1,110.983$. Using the estimate of α_0 from (6.43), the predicted salary is about 1,262,077, or \$1,262,077. Using the estimate from (6.44) gives an estimated salary of about \$1,240,968. These differ from each other by much less than each differs from the naive prediction of \$1,110,983.

We can use the previous method of obtaining predictions to determine how well the model with $\log(y)$ as the dependent variable explains y . We already have measures for models when y is the dependent variable: the R -squared and the adjusted R -squared. The goal is to find a goodness-of-fit measure in the $\log(y)$ model that can be compared with an R -squared from a model where y is the dependent variable.

There are different ways to define a goodness-of-fit measure after retransforming a model for $\log(y)$ to predict y . Here we present an approach that is easy to implement and that gives the same value whether we estimate α_0 as in (6.40), (6.43), or (6.44). To motivate the measure, recall that in the linear regression equation estimated by OLS,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_k x_k, \quad [6.46]$$

the usual R -squared is simply the square of the correlation between y_i and $\hat{y}_i$ (see Section 3.2). Now, if instead we compute fitted values from (6.42)—that is, $\hat{y}_i = \hat{\alpha}_0 m_i$ for all observations i —then it makes sense to use the square of the correlation between y_i and these fitted values as an R -squared. Because correlation is unaffected if we multiply by a constant, it does not matter which estimate of α_0 we use. In fact, this R -squared measure for y [not $\log(y)$] is just the squared correlation between y_i and $\hat{m}_i$. We can compare this directly with the R -squared from equation (6.46). [Because the R -squared calculation does not depend on the estimate of α_0 , it does not allow us to choose among (6.40), (6.43), and (6.44). But we know that (6.44) minimizes the sum of squared residuals between y_i and $\hat{m}_i$, without a constant. In other words, given the $\hat{m}_i$, $\check{\alpha}_0$ is chosen to produce the best fit based on sum of squared residuals. We are interested here in choosing between the linear model for y and $\log(y)$, and so an R -squared measure that does not depend on how we estimate α_0 is suitable.]

EXAMPLE 6.8

PREDICTING CEO SALARIES

After we obtain the $\hat{m}_i$, we just obtain the correlation between $salary_i$ and $\hat{m}_i$; it is .493. The square of it is about .243, and this is a measure of how well the log model explains the variation in $salary$, not $\log(salary)$. [The R^2 from (6.45), .318, tells us that the log model explains about 31.8% of the variation in $\log(salary)$.]

As a competing linear model, suppose we estimate a model with all variables in levels:

$$salary = \beta_0 + \beta_1 sales + \beta_2 mktval + \beta_3 ceoten + u. \quad [6.47]$$

The key is that the dependent variable is $salary$. We could use logs of $sales$ or $mktval$ on the right-hand side, but it makes more sense to have all dollar values in levels if one ($salary$) appears as a level. The R -squared from estimating this equation using the same 177 observations is .201. Thus, the log model explains more of the variation in $salary$, and so we prefer it to (6.47) on goodness-of-fit grounds. The log model is also preferred because it seems more realistic and its parameters are easier to interpret.

If we maintain the full set of classical linear model assumptions in the model (6.38), we can easily obtain prediction intervals for $y^0 = \exp(\beta_0 + \beta_1 x_1^0 + \dots + \beta_k x_k^0 + u^0)$ when we have estimated a linear model for $\log(y)$. Recall that $x_1^0, x_2^0, \dots, x_k^0$ are known values and u^0 is the unobserved error that partly determines y^0 . From equation (6.37), a 95% prediction interval for $\log y^0 = \log(y^0)$ is simply $\widehat{\log y^0} \pm t_{.025} \cdot \text{se}(\hat{e}^0)$, where $\text{se}(\hat{e}^0)$ is obtained from the regression of $\log(y)$ on $x_1, \dots, x_k$ using the original n observations. Let $c_l = -t_{.025} \cdot \text{se}(\hat{e}^0)$ and $c_u = t_{.025} \cdot \text{se}(\hat{e}^0)$ be the lower and upper bounds of the prediction interval for $\log y^0$. That is, $P(c_l \leq \log y^0 \leq c_u) = .95$. Because the exponential function is strictly increasing, it is also true that $P[\exp(c_l) \leq \exp(\log y^0) \leq \exp(c_u)] = .95$, that is, $P[\exp(c_l) \leq y^0 \leq \exp(c_u)] = .95$. Therefore, we can take $\exp(c_l)$ and $\exp(c_u)$ as the lower and upper bounds, respectively, for a 95% prediction interval for y^0 . For large n , $t_{.025} = 1.96$, and so a 95% prediction interval for y^0 is $\exp[-1.96 \cdot \text{se}(\hat{e}^0)] \exp(\hat{\beta}_0 + \mathbf{x}^0 \hat{\boldsymbol{\beta}})$ to $\exp[1.96 \cdot \text{se}(\hat{e}^0)] \exp(\hat{\beta}_0 + \mathbf{x}^0 \hat{\boldsymbol{\beta}})$, where $\mathbf{x}^0 \hat{\boldsymbol{\beta}}$ is shorthand for $\hat{\beta}_1 x_1^0 + \dots + \hat{\beta}_k x_k^0$. Remember, the $\hat{\beta}_j$ and $\text{se}(\hat{e}^0)$ are obtained from the regression with $\log(y)$ as the dependent variable. Because we assume normality of u in (6.38), we probably would use (6.40) to obtain a point prediction for y^0 . Unlike in equation (6.37), this point prediction will not lie halfway between the upper and lower bounds $\exp(c_l)$ and $\exp(c_u)$. One can obtain different 95% prediction intervals by choosing different quantiles in the t_{n-k-1} distribution. If q_{α_1} and q_{α_2} are quantiles with $\alpha_2 - \alpha_1 = .95$, then we can choose $c_l = q_{\alpha_1} \text{se}(\hat{e}^0)$ and $c_u = q_{\alpha_2} \text{se}(\hat{e}^0)$.

As an example, consider the CEO salary regression, where we make the prediction at the same values of $sales$, $mktval$, and $ceoten$ as in Example 6.7. The standard error of the regression for (6.43) is about .505, and the standard error of $\log y^0$ is about .075. Therefore, using equation (6.36), $\text{se}(\hat{e}^0) \approx .511$; as in the GPA example, the error variance swamps the estimation error in the parameters, even though here the sample size is only 177. A 95% prediction interval for $salary^0$ is $\exp[-1.96 \cdot (.511)] \exp(7.013)$ to $\exp[1.96 \cdot (.511)] \exp(7.013)$, or about 408,071 to 3,024,678, that is, \$408,071 to \$3,024,678. This very wide 95% prediction interval for CEO salary at the given sales, market value, and tenure values shows that there is much else that we have not included in the regression that determines salary. Incidentally, the point prediction for salary, using (6.40), is about \$1,262,075—higher than the predictions using the other estimates of α_0 and closer to the lower bound than the upper bound of the 95% prediction interval.

Summary

In this chapter, we have covered some important multiple regression analysis topics.

Section 6.1 showed that a change in the units of measurement of an independent variable changes the OLS coefficient in the expected manner: if x_j is multiplied by c , its coefficient is divided by c . If the dependent variable is multiplied by c , *all* OLS coefficients are multiplied by c . Neither t nor F statistics are affected by changing the units of measurement of any variables.

We discussed beta coefficients, which measure the effects of the independent variables on the dependent variable in standard deviation units. The beta coefficients are obtained from a standard OLS regression after the dependent and independent variables have been transformed into z -scores.

We provided a detailed discussion of functional form, including the logarithmic transformation, quadratics, and interaction terms. It is helpful to summarize some of our conclusions.

CONSIDERATIONS WHEN USING LOGARITHMS

1. The coefficients have percentage change interpretations. We can be ignorant of the units of measurement of any variable that appears in logarithmic form, and changing units from, say, dollars to thousands of dollars has no effect on a variable's coefficient when that variable appears in logarithmic form.
2. Logs are often used for dollar amounts that are always positive, as well as for variables such as population, especially when there is a lot of variation. They are used less often for variables measured in years, such as schooling, age, and experience. Logs are used infrequently for variables that are already percents or proportions, such as an unemployment rate or a pass rate on a test.
3. Models with $\log(y)$ as the dependent variable often more closely satisfy the classical linear model assumptions. For example, the model has a better chance of being linear, homoskedasticity is more likely to hold, and normality is often more plausible.
4. In many cases, taking the log greatly reduces the variation of a variable, making OLS estimates less prone to outlier influence. However, in cases where y is a fraction and close to zero for many observations, $\log(y_i)$ can have much more variability than y_i . For values y_i very close to zero, $\log(y_i)$ is a negative number very large in magnitude.
5. If $y \geq 0$ but $y = 0$ is possible, we cannot use $\log(y)$. Sometimes $\log(1 + y)$ is used, but interpretation of the coefficients is difficult.
6. For large changes in an explanatory variable, we can compute a more accurate estimate of the percentage change effect.
7. It is harder (but possible) to predict y when we have estimated a model for $\log(y)$.

CONSIDERATIONS WHEN USING QUADRATICS

1. A quadratic function in an explanatory variable allows for an increasing or decreasing effect.
2. The turning point of a quadratic is easily calculated, and it should be calculated to see if it makes sense.
3. Quadratic functions where the coefficients have the opposite sign have a strictly positive turning point; if the signs of the coefficients are the same, the turning point is at a negative value of x .
4. A seemingly small coefficient on the square of a variable can be practically important in what it implies about a changing slope. One can use a t test to see if the quadratic is statistically significant, and compute the slope at various values of x to see if it is practically important.

5. For a model quadratic in a variable x , the coefficient on x measures the partial effect starting from $x = 0$, as can be seen in equation (6.11). If zero is not a possible or interesting value of x , one can center x about a more interesting value, such as the average in the sample, before computing the square. Computing Exercise 6.12 provides an example.

CONSIDERATIONS WHEN USING INTERACTIONS

1. Interaction terms allow the partial effect of an explanatory variable, say x_1 , to depend on the level of another variable, say x_2 —and vice versa.
2. Interpreting models with interactions can be tricky. The coefficient on x_1 , say β_1 , measures the partial effect of x_1 on y when $x_2 = 0$, which may be impossible or uninteresting. Centering x_1 and x_2 around interesting values before constructing the interaction term typically leads to an equation that is visually more appealing.
3. A standard t test can be used to determine if an interaction term is statistically significant. Computing the partial effects at different values of the explanatory variables can be used to determine the practical importance of interactions.

We introduced the adjusted R -squared, $\bar{R}^2$, as an alternative to the usual R -squared for measuring goodness-of-fit. Whereas R^2 can never fall when another variable is added to a regression, $\bar{R}^2$ penalizes the number of regressors and can drop when an independent variable is added. This makes $\bar{R}^2$ preferable for choosing between nonnested models with different numbers of explanatory variables. Neither R^2 nor $\bar{R}^2$ can be used to compare models with different dependent variables. Nevertheless, it is fairly easy to obtain goodness-of-fit measures for choosing between y and $\log(y)$ as the dependent variable, as shown in Section 6.4.

In Section 6.3, we discussed the somewhat subtle problem of relying too much on R^2 or $\bar{R}^2$ in arriving at a final model: it is possible to control for too many factors in a regression model. For this reason, it is important to think ahead about model specification, particularly the *ceteris paribus* nature of the multiple regression equation. Explanatory variables that affect y and are uncorrelated with all the other explanatory variables can be used to reduce the error variance without inducing multicollinearity.

In Section 6.4, we demonstrated how to obtain a confidence interval for a prediction made from an OLS regression line. We also showed how a confidence interval can be constructed for a future, unknown value of y .

Occasionally, we want to predict y when $\log(y)$ is used as the dependent variable in a regression model. Section 6.4 explains this simple method. Finally, we are sometimes interested in knowing about the sign and magnitude of the residuals for particular observations. Residual analysis can be used to determine whether particular members of the sample have predicted values that are well above or well below the actual outcomes.

Key Terms

Adjusted R -Squared	Over Controlling	Resampling Method
Beta Coefficients	Population R -Squared	Residual Analysis
Bootstrap	Prediction Error	Smearing Estimate
Bootstrap Standard Error	Prediction Interval	Standardized Coefficients
Interaction Effect	Predictions	Variance of the Prediction
Nonnested Models	Quadratic Functions	Error

Problems

- 1 The following equation was estimated using the data in CEOSAL1.RAW:

$$\widehat{\log(\text{salary})} = 4.322 + .276 \log(\text{sales}) + .0215 \text{roe} - .00008 \text{roe}^2$$

$$(.324) \quad (.033) \quad (.0129) \quad (.00026)$$

$$n = 209, R^2 = .282.$$

This equation allows *roe* to have a diminishing effect on $\log(\text{salary})$. Is this generality necessary? Explain why or why not.

- 2 Let $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ be the OLS estimates from the regression of y_i on $x_{i1}, \dots, x_{ik}$, $i = 1, 2, \dots, n$. For nonzero constants $c_1, \dots, c_k$, argue that the OLS intercept and slopes from the regression of $c_0 y_i$ on $c_1 x_{i1}, \dots, c_k x_{ik}$, $i = 1, 2, \dots, n$, are given by $\hat{\beta}_0 = c_0 \hat{\beta}_0$, $\hat{\beta}_1 = (c_0/c_1) \hat{\beta}_1$, $\dots$, $\hat{\beta}_k = (c_0/c_k) \hat{\beta}_k$. [Hint: Use the fact that the $\hat{\beta}_j$ solve the first order conditions in (3.13), and the $\hat{\beta}_j$ must solve the first order conditions involving the rescaled dependent and independent variables.]

- 3 Using the data in RDCHEM.RAW, the following equation was obtained by OLS:

$$\widehat{\text{rdintens}} = 2.613 + .00030 \text{sales} - .0000000070 \text{sales}^2$$

$$(.429) \quad (.00014) \quad (.0000000037)$$

$$n = 32, R^2 = .1484.$$

- At what point does the marginal effect of *sales* on *rdintens* become negative?
 - Would you keep the quadratic term in the model? Explain.
 - Define *salesbil* as sales measured in billions of dollars: $\text{salesbil} = \text{sales}/1,000$. Rewrite the estimated equation with *salesbil* and salesbil^2 as the independent variables. Be sure to report standard errors and the *R*-squared. [Hint: Note that $\text{salesbil}^2 = \text{sales}^2/(1,000)^2$.]
 - For the purpose of reporting the results, which equation do you prefer?
- 4 The following model allows the return to education to depend upon the total amount of both parents' education, called *pareduc*:

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{educ} \cdot \text{pareduc} + \beta_3 \text{exper} + \beta_4 \text{tenure} + u.$$

- Show that, in decimal form, the return to another year of education in this model is

$$\Delta \log(\text{wage}) / \Delta \text{educ} = \beta_1 + \beta_2 \text{pareduc}.$$

What sign do you expect for β_2 ? Why?

- Using the data in WAGE2.RAW, the estimated equation is

$$\widehat{\log(\text{wage})} = 5.65 + .047 \text{educ} + .00078 \text{educ} \cdot \text{pareduc} +$$

$$(.13) \quad (.010) \quad (.00021)$$

$$.019 \text{exper} + .010 \text{tenure}$$

$$(.004) \quad (.003)$$

$$n = 722, R^2 = .169.$$

(Only 722 observations contain full information on parents' education.) Interpret the coefficient on the interaction term. It might help to choose two specific values for *pareduc*—for example, *pareduc* = 32 if both parents have a college education, or *pareduc* = 24 if both parents have a high school education—and to compare the estimated return to *educ*.

(iii) When *pareduc* is added as a separate variable to the equation, we get:

$$\begin{aligned}\widehat{\log(\text{wage})} &= 4.94 + .097 \text{ educ} + .033 \text{ pareduc} - .0016 \text{ educ} \cdot \text{pareduc} \\ &\quad (.38) \quad (.027) \quad (.017) \quad (.0012) \\ &\quad + .020 \text{ exper} + .010 \text{ tenure} \\ &\quad (.004) \quad (.003) \\ n &= 722, R^2 = .174.\end{aligned}$$

Does the estimated return to education now depend positively on parent education? Test the null hypothesis that the return to education does not depend on parent education.

- 5 In Example 4.2, where the percentage of students receiving a passing score on a tenth-grade math exam (*math10*) is the dependent variable, does it make sense to include *scil1*—the percentage of eleventh graders passing a science exam—as an additional explanatory variable?
- 6 When *atndrte*² and *ACT·atndrte* are added to the equation estimated in (6.19), the *R*-squared becomes .232. Are these additional terms jointly significant at the 10% level? Would you include them in the model?
- 7 The following three equations were estimated using the 1,534 observations in 401K.RAW:

$$\begin{aligned}\widehat{\text{prate}} &= 80.29 + 5.44 \text{ mrate} + .269 \text{ age} - .00013 \text{ totemp} \\ &\quad (.78) \quad (.52) \quad (.045) \quad (.00004) \\ R^2 &= .100, \bar{R}^2 = .098.\end{aligned}$$

$$\begin{aligned}\widehat{\text{prate}} &= 97.32 + 5.02 \text{ mrate} + .314 \text{ age} - 2.66 \log(\text{totemp}) \\ &\quad (1.95) \quad (.051) \quad (.044) \quad (.28) \\ R^2 &= .144, \bar{R}^2 = .142.\end{aligned}$$

$$\begin{aligned}\widehat{\text{prate}} &= 80.62 + 5.34 \text{ mrate} + .290 \text{ age} - .00043 \text{ totemp} \\ &\quad (.78) \quad (.52) \quad (.045) \quad (.00009) \\ &\quad + .0000000039 \text{ totemp}^2 \\ &\quad (.0000000010) \\ R^2 &= .108, \bar{R}^2 = .106.\end{aligned}$$

Which of these three models do you prefer? Why?

- 8 Suppose we want to estimate the effects of alcohol consumption (*alcohol*) on college grade point average (*colGPA*). In addition to collecting information on grade point averages and alcohol usage, we also obtain attendance information (say, percentage of lectures attended, called *attend*). A standardized test score (say, *SAT*) and high school GPA (*hsGPA*) are also available.
 - (i) Should we include *attend* along with *alcohol* as explanatory variables in a multiple regression model? (Think about how you would interpret β_{alcohol} .)
 - (ii) Should *SAT* and *hsGPA* be included as explanatory variables? Explain.

- 9 If we start with (6.38) under the CLM assumptions, assume large n , and ignore the estimation error in the $\hat{\beta}_j$, a 95% prediction interval for y^0 is $[\exp(-1.96\hat{\sigma}) \exp(\widehat{\log y^0}), \exp(1.96\hat{\sigma}) \exp(\widehat{\log y^0})]$. The point prediction for y^0 is $\hat{y}^0 = \exp(\hat{\sigma}^2/2) \exp(\widehat{\log y^0})$.
- (i) For what values of $\hat{\sigma}$ will the point prediction be in the 95% prediction interval? Does this condition seem likely to hold in most applications?
 - (ii) Verify that the condition from part (i) is satisfied in the CEO salary example.

Computer Exercises

- C1 Use the data in KIELMC.RAW, only for the year 1981, to answer the following questions. The data are for houses that sold during 1981 in North Andover, Massachusetts; 1981 was the year construction began on a local garbage incinerator.
- (i) To study the effects of the incinerator location on housing price, consider the simple regression model

$$\log(\text{price}) = \beta_0 + \beta_1 \log(\text{dist}) + u,$$

where *price* is housing price in dollars and *dist* is distance from the house to the incinerator measured in feet. Interpreting this equation causally, what sign do you expect for β_1 if the presence of the incinerator depresses housing prices? Estimate this equation and interpret the results.

- (ii) To the simple regression model in part (i), add the variables $\log(\text{intst})$, $\log(\text{area})$, $\log(\text{land})$, *rooms*, *baths*, and *age*, where *intst* is distance from the home to the interstate, *area* is square footage of the house, *land* is the lot size in square feet, *rooms* is total number of rooms, *baths* is number of bathrooms, and *age* is age of the house in years. Now, what do you conclude about the effects of the incinerator? Explain why (i) and (ii) give conflicting results.
- (iii) Add $[\log(\text{intst})]^2$ to the model from part (ii). Now what happens? What do you conclude about the importance of functional form?
- (iv) Is the square of $\log(\text{dist})$ significant when you add it to the model from part (iii)?

- C2 Use the data in WAGE1.RAW for this exercise.

- (i) Use OLS to estimate the equation

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + u$$

and report the results using the usual format.

- (ii) Is exper^2 statistically significant at the 1% level?
- (iii) Using the approximation

$$\% \Delta \widehat{\text{wage}} \approx 100(\hat{\beta}_2 + 2\hat{\beta}_3 \text{exper}) \Delta \text{exper},$$

find the approximate return to the fifth year of experience. What is the approximate return to the twentieth year of experience?

- (iv) At what value of *exper* does additional experience actually lower predicted $\log(\text{wage})$? How many people have more experience in this sample?

- C3 Consider a model where the return to education depends upon the amount of work experience (and vice versa):

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{educ} \cdot \text{exper} + u.$$

- (i) Show that the return to another year of education (in decimal form), holding *exper* fixed, is $\beta_1 + \beta_3 \text{exper}$.
- (ii) State the null hypothesis that the return to education does not depend on the level of *exper*. What do you think is the appropriate alternative?
- (iii) Use the data in WAGE2.RAW to test the null hypothesis in (ii) against your stated alternative.
- (iv) Let θ_1 denote the return to education (in decimal form), when *exper* = 10: $\theta_1 = \beta_1 + 10\beta_3$. Obtain $\hat{\theta}_1$ and a 95% confidence interval for θ_1 . (Hint: Write $\beta_1 = \theta_1 - 10\beta_3$ and plug this into the equation; then rearrange. This gives the regression for obtaining the confidence interval for θ_1 .)

C4 Use the data in GPA2.RAW for this exercise.

- (i) Estimate the model

$$\text{sat} = \beta_0 + \beta_1 \text{hsize} + \beta_2 \text{hsize}^2 + u,$$

where *hsize* is the size of the graduating class (in hundreds), and write the results in the usual form. Is the quadratic term statistically significant?

- (ii) Using the estimated equation from part (i), what is the “optimal” high school size? Justify your answer.
- (iii) Is this analysis representative of the academic performance of *all* high school seniors? Explain.
- (iv) Find the estimated optimal high school size, using $\log(\text{sat})$ as the dependent variable. Is it much different from what you obtained in part (ii)?

C5 Use the housing price data in HPRICE1.RAW for this exercise.

- (i) Estimate the model

$$\log(\text{price}) = \beta_0 + \beta_1 \log(\text{lotsize}) + \beta_2 \log(\text{sqrft}) + \beta_3 \text{bdrms} + u$$

and report the results in the usual OLS format.

- (ii) Find the predicted value of $\log(\text{price})$, when *lotsize* = 20,000, *sqrft* = 2,500, and *bdrms* = 4. Using the methods in Section 6.4, find the predicted value of *price* at the same values of the explanatory variables.
- (iii) For explaining variation in *price*, decide whether you prefer the model from part (i) or the model

$$\text{price} = \beta_0 + \beta_1 \text{lotsize} + \beta_2 \text{sqrft} + \beta_3 \text{bdrms} + u.$$

C6 Use the data in VOTE1.RAW for this exercise.

- (i) Consider a model with an interaction between expenditures:

$$\text{voteA} = \beta_0 + \beta_1 \text{prtystrA} + \beta_2 \text{expendA} + \beta_3 \text{expendB} + \beta_4 \text{expendA} \cdot \text{expendB} + u.$$

What is the partial effect of *expendB* on *voteA*, holding *prtystrA* and *expendA* fixed? What is the partial effect of *expendA* on *voteA*? Is the expected sign for β_4 obvious?

- (ii) Estimate the equation in part (i) and report the results in the usual form. Is the interaction term statistically significant?

- (iii) Find the average of *expendA* in the sample. Fix *expendA* at 300 (for \$300,000). What is the estimated effect of another \$100,000 spent by Candidate B on *voteA*? Is this a large effect?
- (iv) Now fix *expendB* at 100. What is the estimated effect of $\Delta \text{expendA} = 100$ on *voteA*? Does this make sense?
- (v) Now, estimate a model that replaces the interaction with *shareA*, Candidate A's percentage share of total campaign expenditures. Does it make sense to hold both *expendA* and *expendB* fixed, while changing *shareA*?
- (vi) (Requires calculus) In the model from part (v), find the partial effect of *expendB* on *voteA*, holding *prtystrA* and *expendA* fixed. Evaluate this at *expendA* = 300 and *expendB* = 0 and comment on the results.

C7 Use the data in ATTEND.RAW for this exercise.

- (i) In the model of Example 6.3, argue that

$$\Delta \text{stdfnl} / \Delta \text{priGPA} \approx \beta_2 + 2\beta_4 \text{priGPA} + \beta_6 \text{atndrte}.$$

Use equation (6.19) to estimate the partial effect when *priGPA* = 2.59 and *atndrte* = 82. Interpret your estimate.

- (ii) Show that the equation can be written as

$$\begin{aligned} \text{stdfnl} = & \theta_0 + \beta_1 \text{atndrte} + \theta_2 \text{priGPA} + \beta_3 \text{ACT} + \beta_4 (\text{priGPA} - 2.59)^2 \\ & + \beta_5 \text{ACT}^2 + \beta_6 \text{priGPA} (\text{atndrte} - 82) + u, \end{aligned}$$

where $\theta_2 = \beta_2 + 2\beta_4(2.59) + \beta_6(82)$. (Note that the intercept has changed, but this is unimportant.) Use this to obtain the standard error of $\hat{\theta}_2$ from part (i).

- (iii) Suppose that, in place of *priGPA*(*atndrte* - 82), you put (*priGPA* - 2.59) · (*atndrte* - 82). Now how do you interpret the coefficients on *atndrte* and *priGPA*?

C8 Use the data in HPRICE1.RAW for this exercise.

- (i) Estimate the model

$$\text{price} = \beta_0 + \beta_1 \text{lotsize} + \beta_2 \text{sqrft} + \beta_3 \text{bdrms} + u$$

and report the results in the usual form, including the standard error of the regression. Obtain predicted price, when we plug in *lotsize* = 10,000, *sqrft* = 2,300, and *bdrms* = 4; round this price to the nearest dollar.

- (ii) Run a regression that allows you to put a 95% confidence interval around the predicted value in part (i). Note that your prediction will differ somewhat due to rounding error.
- (iii) Let price^0 be the unknown future selling price of the house with the characteristics used in parts (i) and (ii). Find a 95% CI for price^0 and comment on the width of this confidence interval.

C9 The data set NBASAL.RAW contains salary information and career statistics for 269 players in the National Basketball Association (NBA).

- (i) Estimate a model relating points-per-game (*points*) to years in the league (*exper*), *age*, and years played in college (*coll*). Include a quadratic in *exper*; the other variables should appear in level form. Report the results in the usual way.
- (ii) Holding college years and age fixed, at what value of experience does the next year of experience actually reduce points-per-game? Does this make sense?

- (iii) Why do you think *coll* has a negative and statistically significant coefficient? (*Hint*: NBA players can be drafted before finishing their college careers and even directly out of high school.)
- (iv) Add a quadratic in *age* to the equation. Is it needed? What does this appear to imply about the effects of age, once experience and education are controlled for?
- (v) Now regress $\log(\text{wage})$ on *points*, *exper*, exper^2 , *age*, and *coll*. Report the results in the usual format.
- (vi) Test whether *age* and *coll* are jointly significant in the regression from part (v). What does this imply about whether age and education have separate effects on wage, once productivity and seniority are accounted for?

C10 Use the data in BWGHT2.RAW for this exercise.

- (i) Estimate the equation

$$\log(\text{bwght}) = \beta_0 + \beta_1 \text{npvis} + \beta_2 \text{npvis}^2 + u$$

- by OLS, and report the results in the usual way. Is the quadratic term significant?
- (ii) Show that, based on the equation from part (i), the number of prenatal visits that maximizes $\log(\text{bwght})$ is estimated to be about 22. How many women had at least 22 prenatal visits in the sample?
- (iii) Does it make sense that birth weight is actually predicted to decline after 22 prenatal visits? Explain.
- (iv) Add mother's age to the equation, using a quadratic functional form. Holding *npvis* fixed, at what mother's age is the birth weight of the child maximized? What fraction of women in the sample are older than the "optimal" age?
- (v) Would you say that mother's age and number of prenatal visits explain a lot of the variation in $\log(\text{bwght})$?
- (vi) Using quadratics for both *npvis* and *age*, decide whether using the natural log or the level of *bwght* is better for predicting *bwght*.

C11 Use APPLE.RAW to verify some of the claims made in Section 6.3.

- (i) Run the regression *ecolbs* on *ecoprc*, *regprc* and report the results in the usual form, including the *R*-squared and adjusted *R*-squared. Interpret the coefficients on the price variables and comment on their signs and magnitudes.
- (ii) Are the price variables statistically significant? Report the *p*-values for the individual *t* tests.
- (iii) What is the range of fitted values for *ecolbs*? What fraction of the sample reports *ecolbs* = 0? Comment.
- (iv) Do you think the price variables together do a good job of explaining variation in *ecolbs*? Explain.
- (v) Add the variables *faminc*, *hhsize* (household size), *educ*, and *age* to the regression from part (i). Find the *p*-value for their joint significance. What do you conclude?
- (vi) Run separate simple regressions of *ecolbs* on *ecoprc* and then *ecolbs* on *regprc*. How do the simple regression coefficients compare with the multiple regression from part (i)? Find the correlation coefficient between *ecoprc* and *regprc* to help explain your findings.

C12 Use the subset of 401KSUBS.RAW with *fsize* = 1; this restricts the analysis to single-person households; see also Computer Exercise C8 in Chapter 4.

- (i) What is the youngest age of people in this sample? How many people are at that age?

- (ii) In the model

$$nettfa = \beta_0 + \beta_1 inc + \beta_2 age + \beta_3 age^2 + u,$$

what is the literal interpretation of β_2 ? By itself, is it of much interest?

- (iii) Estimate the model from part (ii) and report the results in standard form. Are you concerned that the coefficient on *age* is negative? Explain.
- (iv) Because the youngest people in the sample are 25, it makes sense to think that, for a given level of income, the lowest average amount of net total financial assets is at age 25. Recall that the partial effect of *age* on *nettfa* is $\beta_2 + 2\beta_3 age$, so the partial effect at age 25 is $\beta_2 + 2\beta_3(25) = \beta_2 + 50\beta_3$; call this θ_2 . Find $\hat{\theta}_2$ and obtain the two-sided *p*-value for testing $H_0: \theta_2 = 0$. You should conclude that $\hat{\theta}_2$ is small and very statistically insignificant. [Hint: One way to do this is to estimate the model $nettfa = \alpha_0 + \beta_1 inc + \theta_2 age + \beta_3(age - 25)^2 + u$, where the intercept, α_0 , is different from β_0 . There are other ways, too.]
- (v) Because the evidence against $H_0: \theta_2 = 0$ is very weak, set it to zero and estimate the model

$$nettfa = \alpha_0 + \beta_1 inc + \beta_3(age - 25)^2 + u.$$

In terms of goodness-of-fit, does this model fit better than that in part (ii)?

- (vi) For the estimated equation in part (v), set *inc* = 30 (roughly, the average value) and graph the relationship between *nettfa* and *age*, but only for *age* ≥ 25 . Describe what you see.
- (vii) Check to see whether including a quadratic in *inc* is necessary.

C13 Use the data in MEAP00.RAW to answer this question.

- (i) Estimate the model

$$math4 = \beta_0 + \beta_2 lexppp + \beta_2 lenroll + \beta_3 lunch + u$$

by OLS, and report the results in the usual form. Is each explanatory variable statistically significant at the 5% level?

- (ii) Obtain the fitted values from the regression in part (i). What is the range of fitted values? How does it compare with the range of the actual data on *math4*?
- (iii) Obtain the residuals from the regression in part (i). What is the building code of the school that has the largest (positive) residual? Provide an interpretation of this residual.
- (iv) Add quadratics of all explanatory variables to the equation, and test them for joint significance. Would you leave them in the model?
- (v) Returning to the model in part (i), divide the dependent variable and each explanatory variable by its sample standard deviation, and rerun the regression. (Include an intercept unless you also first subtract the mean from each variable.) In terms of standard deviation units, which explanatory variable has the largest effect on the math pass rate?

C14 Use the data in BENEFITS.RAW to answer this question. It is a school-level data set at the K–5 level on average teacher salary and benefits. See Example 4.10 for background.

- (i) Regress *lavgsal* on *bs* and report the results in the usual form. Can you reject $H_0: \beta_{bs} = 0$ against a two-sided alternative? Can you reject $H_0: \beta_{bs} = -1$ against $H_1: \beta_{bs} > -1$? Report the *p*-values for both tests.

- (ii) Define $lbs = \log(bs)$. Find the range of values for lbs and find its standard deviation. How do these compare to the range and standard deviation for bs ?
- (iii) Regress $lavg\text{sal}$ on lbs . Does this fit better than the regression from part (i)?
- (iv) Estimate the equation

$$lavg\text{sal} = \beta_0 + \beta_1 lbs + \beta_2 lenroll + \beta_3 lstaff + \beta_4 lunch + u$$
 and report the results in the usual form. What happens to the coefficient on bs ? Is it now statistically different from zero?
- (v) Interpret the coefficient on $lstaff$. Why do you think it is negative?
- (vi) Add $lunch^2$ to the equation from part (iv). Is it statistically significant? Compute the turning point (minimum value) in the quadratic, and show that it is within the range of the observed data on $lunch$. How many values of $lunch$ are higher than the calculated turning point?
- (vii) Based on the findings from part (vi), describe how teacher salaries relate to school poverty rates. In terms of teacher salary, and holding other factors fixed, is it better to teach at a school with $lunch = 0$ (no poverty), $lunch = 50$, or $lunch = 100$ (all kids eligible for the free lunch program)?

APPENDIX 6A

6A. A Brief Introduction to Bootstrapping

In many cases where formulas for standard errors are hard to obtain mathematically, or where they are thought not to be very good approximations to the true sampling variation of an estimator, we can rely on a **resampling method**. The general idea is to treat the observed data as a population that we can draw samples from. The most common resampling method is the **bootstrap**. (There are actually several versions of the bootstrap, but the most general, and most easily applied, is called the *nonparametric bootstrap*, and that is what we describe here.)

Suppose we have an estimate, $\hat{\theta}$, of a population parameter, θ . We obtained this estimate, which could be a function of OLS estimates (or estimates that we cover in later chapters), from a random sample of size n . We would like to obtain a standard error for $\hat{\theta}$ that can be used for constructing t statistics or confidence intervals. Remarkably, we can obtain a valid standard error by computing the estimate from different random samples drawn from the original data.

Implementation is easy. If we list our observations from 1 through n , we draw n numbers randomly, with replacement, from this list. This produces a new data set (of size n) that consists of the original data, but with many observations appearing multiple times (except in the rather unusual case that we resample the original data). Each time we randomly sample from the original data, we can estimate θ using the same procedure that we used on the original data. Let $\hat{\theta}^{(b)}$ denote the estimate from bootstrap sample b . Now, if we repeat the resampling and estimation m times, we have m new estimates, $\{\hat{\theta}^{(b)}: b = 1, 2, \dots, m\}$. The **bootstrap standard error** of $\hat{\theta}$ is just the sample standard deviation of the $\hat{\theta}^{(b)}$, namely,

$$\text{bse}(\hat{\theta}) = \left[(m - 1)^{-1} \sum_{b=1}^m (\hat{\theta}^{(b)} - \bar{\hat{\theta}})^2 \right]^{1/2}, \quad [6.48]$$

where $\bar{\hat{\theta}}$ is the average of the bootstrap estimates.

If obtaining an estimate of θ on a sample of size n requires little computational time, as in the case of OLS and all the other estimators we encounter in this text, we can afford to choose m —the number of bootstrap replications—to be large. A typical value is $m = 1,000$, but even $m = 500$ or a somewhat smaller value can produce a reliable standard error. Note that the size of m —the number of times we resample the original data—has nothing to do with the sample size, n . (For certain estimation problems beyond the scope of this text, a large n can force one to do fewer bootstrap replications.) Many statistics and econometrics packages have built-in bootstrap commands, and this makes the calculation of bootstrap standard errors simple, especially compared with the work often required to obtain an analytical formula for an asymptotic standard error.

One can actually do better in most cases by using the bootstrap sample to compute p -values for t statistics (and F statistics), or for obtaining confidence intervals, rather than obtaining a bootstrap standard error to be used in the construction of t statistics or confidence intervals. See Horowitz (2001) for a comprehensive treatment.